The Regulatory Intermediation Gap: Evidence from Singapore

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Abstract

When government equity-market interventions operate through private intermediaries, intermediary constraints can favour firms that already meet selection criteria over the intended beneficiaries. We term this divergence the 'regulatory intermediation gap'. A cross-sectional event study of Singapore's 2025 equity-market reform (a S$5 billion capital deployment channelled through professional fund managers and targeting small- and mid-cap firms on the Singapore Exchange (SGX), bundled with deregulation of SGX's post-listing oversight regime) documents the gap along two dimensions. On the capital-deployment side, the smallest quartile of firms lost 2.5 percent at the bundled announcement while the largest gained 0.2 percent, consistent with fund managers' investability constraints. The size gradient reversed when the Monetary Authority of Singapore (MAS) later disclosed mandate terms requiring small- and mid-cap allocation. On the deregulation side, high-friction firms were penalised as the market read oversight removal as information loss, while watch-listed firms received a stigma-relief premium, reinforced when SGX subsequently abolished the watch-list. The findings suggest that mandate specificity and targeted burden removal shape whether intermediated equity programmes reach their intended beneficiaries.

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1 Introduction

In February 2025, Singapore announced one of the most ambitious equity-market reform packages in its history, and the regulatory architecture that produced it is itself unusual. The Monetary Authority of Singapore (MAS) is an integrated regulator charged with developing Singapore as a financial centre alongside its prudential and market-conduct responsibilities, while front-line supervision of the listed market is delegated to Singapore Exchange Regulation (SGX RegCo), the exchange's regulatory subsidiary. The reform drew on both sides of that architecture at once. On the developmental side, MAS committed S$5 billion through the Equity Market Development Programme (EQDP) to professional fund managers investing in Singapore Exchange (SGX) equities, with the stated aim of drawing institutional capital toward the small- and mid-cap firms that had long suffered thin trading and depressed valuations. On the supervisory side, SGX RegCo proposed to liberalise the post-listing oversight regime – removing a public financial watch-list, consolidating listing-review functions, and streamlining prospectus and admission requirements. The package therefore paired a capital-deployment instrument with an oversight-deregulation instrument aimed at reviving small- and mid-cap listings, and the months that followed disclosed them in sequence.

Singapore's reform instantiates a general regulatory-design problem that recurs across jurisdictions, especially outside the United States, where benchmarked funds hold disproportionately more large-cap stock and private securities litigation is a weaker substitute for public oversight. Governments confronting the decline of small-cap public equity have turned to two families of intervention: routing public or central-bank capital into listed equities through private intermediaries,[1] and liberalising the listing and oversight rules that govern firms once they are public.[2] Each intervention must pass through an institutional channel before it reaches a firm, and the channel carries its own logic. Capital routed through professional managers follows those managers' investability constraints, and oversight liberalisation withdraws whatever information the oversight regime had been producing. Where the channel's logic diverges from the policy's intended beneficiaries, the intervention can miss them even when it is generously funded. We label this divergence 'the regulatory intermediation gap'.

Two such channels operate in Singapore's reform, and our empirical work proxies each. The capital channel, on the capital-deployment side, runs through the professional fund managers whose mandates, fee structures, and benchmark-relative evaluation tend to favour large, liquid, index-proximate securities, so public money routed through them gravitates toward firms that already satisfy investability criteria rather than toward the smaller firms the programme names. The information channel, on the deregulation side, runs through the public oversight instruments – the queries, halts, suspensions, and visible classifications that investors use to price firms whose quality is hard to verify independently – so withdrawing those instruments removes information the market had been relying on. We proxy the first channel by firm size and the second by a

[1] See, e.g., Korea's Corporate Value-up Programme and the Bank of Japan's purchases of exchange-traded funds, the latter channelling central-bank capital into index-linked funds.

[2] See, e.g., the United Kingdom's overhaul of its Listing Rules and the European Union's Listing Act, which ease the regulatory burden of being listed.

firm's accumulated record of regulatory friction. Because the reform disclosed its components in sequence, the cross-section can price each channel in turn rather than only in combination.

Singapore's reform offers an unusual opportunity to observe the gap, because the integrated architecture allowed the two instruments to be announced together in February and then unbundled over the following nine months. We use a cross-sectional event study of announcement returns to ask how the market priced the capital-deployment and oversight-information channels, and which design features changed that pricing. Since the February package bundled the two instruments while later events activated narrower subsets, announcement returns trace the differential cross-sectional pricing associated with each function's instruments across a common set of firms. We observe the market's valuation of each function's instruments, that is, how developmental and supervisory interventions are priced – not the regulator's deliberations, and so the evidence does not speak to whether the combination of functions within one regulatory architecture shaped the package's design. Our sequence of cross-event tests, placebo comparisons, and robustness checks also supports reading the cross-sectional patterns as the pricing of these two channels rather than as a generic size rotation or deregulation premium.

Our findings indicate that the market first priced the capital programme as a large-cap event. At the bundled February announcement, the smallest quartile of firms earned cumulative abnormal returns of −2.5 percent while the largest gained 0.2 percent, a spread of 2.8 percentage points that runs directly counter to the programme's stated purpose. The pattern is what one would expect when S$5 billion must flow through managers whose fiduciary obligations, fee structures, liquidity screens, and benchmark-relative evaluation favour large, liquid, index-proximate securities. That reading is confirmed when the terms changed. In July, MAS named the first three EQDP managers, deployed an initial S$1.1 billion, and disclosed mandate terms specifying significant allocation to small- and mid-cap equities; the size gradient reversed, with smaller firms now gaining more than larger ones, and the average firm earned a positive abnormal return of roughly three percent. Cumulated across the two events the gap narrowed without fully closing, so mandate specificity offset the intermediation bias rather than eliminating it. The capital channel became effective for the intended segment only once the mandate terms were specific enough to constrain the intermediaries' portfolio tilt.

With regard to the information channel, we find that the market priced the deregulation as information loss for the firms most exposed to oversight. Firms carrying accumulated regulatory friction – those that had drawn prior SGX queries, trading halts, and suspensions – earned lower announcement returns, with each additional friction event predicting roughly a third of a percentage point of decline even after conditioning on size. Investors read the proposed withdrawal of SGX's monitoring apparatus as the removal of information they had been using to price firms whose quality is difficult to verify independently, recreating the market-for-lemons problem that disclosure and monitoring regimes exist to suppress.[3] Decomposing the friction measure sharpens the reading: the penalty is driven by halts and suspensions rather than routine queries, and by older accumulated friction rather than recent scrutiny, which indicates that the market treated regulatory attention as a stock of reputation rather than a flow of current monitoring. Once the

[3]George A Akerlof, 'The Market for "Lemons": Quality Uncertainty and the Market Mechanism' (1970) 84 Q J Econ 488

public query regime is withdrawn, investors can no longer tell which firms the regulator has flagged, and an adverse-selection problem reappears across the previously monitored segment.

The deregulation was not priced uniformly, and the exception locates the design margin on the information channel. Conditional on a firm's overall friction, those on SGX's financial watch-list earned positive announcement returns, and when SGX RegCo abolished the watch-list in October, the previously watch-listed firms gained sharply relative to others. The watch-list label imposed financing and counterparty costs that exceeded its informational content, so its removal was priced as stigma relief even as the removal of genuinely informative monitoring was priced as a loss. The two responses are consistent once the operative design choice is understood to be which instrument is withdrawn rather than how much oversight is removed in aggregate. The watch-list evidence rests on a small number of treated firms and is best read as corroborating the friction results, which carry the analytical weight on this channel.

Taken together, the findings point to a single design lesson: design specificity, rather than the scale or scope of the commitment, is the operative policy variable. When public capital is channelled through professional intermediaries, it follows intermediary incentives unless the mandate is specific enough to redirect them; and when oversight is liberalised as a broad reduction in burden, it removes useful information unless the burden removal is targeted. The July mandate disclosure narrowed the gap on the capital channel; on the information channel, the market separated the targeted removal of a stigmatising label, which it priced as relief, from the broad oversight rollback proposed in February, which it priced as loss.

The paper makes three contributions. First, it documents the regulatory intermediation gap in public equity markets, extending to listed equities a concern previously studied mainly in directed lending and government procurement, where the agents that deploy public capital shape who receives it. Second, it shows that the gap responds to regulatory design, since the cross-sectional reversal between the February announcement and the July mandate disclosure indicates that specifying mandate terms changes how intermediated capital is priced for the target segment. Third, it brings the informational theory of securities regulation to bear on deregulation, showing that the market distinguishes the withdrawal of informative monitoring, which it penalises, from the removal of a stigmatising label, which it rewards. For regulators designing comparable programmes, the implication is that the terms of delegation and the selectivity of burden removal generally govern whether the intervention reaches the firms it targets. The reform also offers useful identifying variation, because the bundled February announcement and its staggered unbundling price the same firms first under the combined package and then as individual instruments are disclosed. That separation across the later events is only partial, so the design rests mainly on cross-sectional decomposition within the February announcement and on the February-to-July reversal, with the boundaries of that strategy set out in Section 4.

The remainder of the paper proceeds as follows. Section 2 develops the intermediation gap as a regulatory-design problem and locates the two channels in the relevant literatures. Section 3 describes Singapore's equity market, the integrated regulatory architecture, and the reform timeline. Section 4 sets out the data and the empirical approach, including the boundaries of the design. Section 5 presents the findings on the two channels and their cross-event behaviour.

Section 6 draws out the implications for regulatory design, and Section 7 concludes.

2 The Intermediation Gap as a Regulatory Design Problem

Although a government that wishes to support an underserved equity segment can describe its objective with some precision, it seldom transacts in the market itself; instead it routes public capital through private fund managers and adjusts the oversight regime that governs how firms are monitored once listed. The regulatory intermediation gap that results is a problem of institutional design rather than of policy ambition, because each delegation introduces an intermediary whose own constraints may diverge from the policy's purpose. The capital channel runs through professional managers whose mandates, fee structures, and benchmark-relative evaluation favour large and liquid securities, so public money tends toward firms that already meet investability criteria. The oversight channel runs through the public instruments – queries, halts, suspensions, and visible classifications – that investors use to price firms whose quality is difficult to verify, so removing those instruments withdraws information as well as burden. Put differently, the gap on each channel turns on how specifically the intervention is designed: this section develops the two channels as design problems with identifiable parameters and frames the question the reform makes observable. The two subsections that follow correspond to the capital channel and the information channel.

2.1 The Capital Channel

Government capital deployed through private agents reaches its intended targets only when the agents' incentives are aligned with the policy. The United States Paycheck Protection Programme supplies the clearest evidence: Granja et al. show that bank-level heterogeneity in disbursement caused funds to flow initially to regions less affected by the pandemic,[4] so that transmission depended on the very institutions delegated to carry it. The lesson generalises beyond crisis lending. The United States Community Reinvestment Act 1977 illustrates how directed-credit obligations can miss their target even without bad faith: enacted to encourage banks to lend in low-income communities in exchange for favourable assessments that eased merger and expansion approvals, it induced institutions to optimise toward the assessment criteria rather than the statutory goal, with lending concentrated around examination dates, grade inflation that left roughly 96 percent of banks rated satisfactory or better, and little evidence that credit shifted toward the lower-income communities the statute named.[5][6][7] Both the agent through which public

[4] João Granja and others, 'Did the paycheck protection program hit the target?' (2022) 145 J Fin Econ 725

[5] Sumit Agarwal, Efraim Benmelech, Nittai Bergman and Amit Seru, 'Did the Community Reinvestment Act (CRA) Lead to Risky Lending?' (NBER Working Paper No 18609, 2012).

[6] Peter Conti-Brown and Brian D Feinstein, 'Banking on a Curve: How to Restore the Community Reinvestment Act' (2023) 13 Harv Bus L Rev 335.

[7] Robert B Avery, Paul S Calem and Glenn B Canner, 'The Effects of the Community Reinvestment Act on Local Communities' (Federal Reserve Board, Division of Research and Statistics, March 2003); Michelle Minton, 'The

capital passes and the terms on which it passes govern whether policy intent survives transmission.

For equity markets specifically, the intermediation chain has been the subject of a sustained policy critique. The Kay review of UK equity markets diagnosed how the proliferation of intermediation had misaligned incentives along the investment chain and encouraged participants to evaluate market performance through the lens of intermediaries rather than of companies or end investors.[8] That review built on earlier official assessments reaching complementary conclusions: the Myners review documented misaligned incentives in institutional investment decision making,[9] while the Walker review identified governance failures in the financial institutions that sit between savers and firms.[10] Braun and Christophers have since characterised the mechanism these reviews describe as 'asset manager capitalism',[11] a label denoting the structural dominance of intermediary constraints over end-investor interests in markets where holdings are concentrated among a small number of large managers. At the level of the largest index managers, Bebchuk and Hirst document the substantial and growing voting power of the dominant fund families together with the agency distortions that delegation produces,[12] which take the form of under-investment in stewardship and deference to portfolio-company management.[13] The supporting asset-pricing theory is well developed. Because delegated managers are evaluated against a benchmark, the optimal contract under unobservable skill induces them to track the index closely,[14] and equilibrium models in which a benchmarked sector coexists with unconstrained investors show that this incentive raises the relative prices of index constituents and dampens the demand that would otherwise flow to securities outside the benchmark.[15] The growing scale of benchmarked capital amplifies the effect, distorting the cost of capital across firms according to their index weight rather than their fundamentals,[16] a pattern visible even among the long-horizon institutions whose mandates ostensibly free them to pursue idiosyncratic value.[17] The reform we study makes this constraint observable as a price: the market's reaction to a S$5 billion programme channelled through fund managers reflects its expectation of where intermediary incentives will direct the capital.

These constraints operate against a structural decline in small-cap public equity. The count of listed firms has fallen across developed markets and has roughly halved in the United States

Community Reinvestment Act's Harmful Legacy: How It Hampers Access to Credit' (Competitive Enterprise Institute, No 132, March 2008).

[8] John Kay, 'The Kay Review of UK Equity Markets and Long-Term Decision Making' (Department for Business, Innovation and Skills 2012)

[9] Paul Myners, 'Institutional Investment in the United Kingdom: A Review' (HM Treasury 2001)

[10] David Walker, 'A Review of Corporate Governance in UK Banks and Other Financial Industry Entities' (HM Treasury 2009)

[11] Benjamin Braun and Brett Christophers, 'Asset Manager Capitalism: An Introduction to Its Political Economy and Economic Geography' (2024) 56 Env & Plan A 546

[12] Lucian Bebchuk and Scott Hirst, 'Index Funds and the Future of Corporate Governance: Theory, Evidence, and Policy' (2019) 119 Colum L Rev 2029

[13] See also Bebchuk and Hirst, 'Big three power, and why it matters' (2022) 102 BU L Rev 1547

[14] Zhiguo He and Wei Xiong, 'Delegated Asset Management, Investment Mandates, and Capital Immobility' (2013) 107 J Fin Econ 239

[15] Andrea Buffa, Dimitri Vayanos and Paul Woolley, 'Asset Management Contracts and Equilibrium Prices' (Working Paper, 2014)

[16] Anil K Kashyap and others, 'The benchmark inclusion subsidy' (2021) 142 J Fin Econ 756

[17] Aleksandar Andonov, 'Delegated Investment Management in Alternative Assets' (2022) 13 Rev Corp Fin Stud 264

from its late-1990s peak,[18] driven substantially by the collapse of small-company flotation.[19] The firms that remain are increasingly concentrated, since rising industry concentration and the disappearance of smaller public competitors coincide with a growing share of institutional capital held in the largest names.[20] The growth of index-linked investing reinforces this pattern at the level of price formation: index demand detaches prices from firm-specific information and disproportionately inflates mega-cap valuations,[21][22] while institutional overweighting of benchmark-proximate holdings, sharpest among managers who track their benchmarks most closely, makes the tilt toward large firms a rational response rather than an anomaly.[23][24] A public programme that targets the small-cap segment therefore works against the prevailing direction of intermediated flows, which sharpens the design question of whether its mandate is specific enough to redirect them.

Governments across several jurisdictions have responded to comparable concerns amid intensifying competition among exchanges for a shrinking pool of listings,[25] and the comparison clarifies what is distinctive about the design margin we isolate. These responses divide by the channel they engage and by how precisely they target an underserved segment. Public-private capital initiatives such as China's Government Guidance Funds route state money through appointed managers, yet they operate predominantly in venture-capital and private-equity channels rather than through secondary-market purchases of listed equities,[26] so the intermediary constraints they confront differ from those binding a listed-equity mandate. Supply-side reforms instead lower the barriers to listing: the European Union's Listing Act, in force from December 2024, and the United Kingdom's 2024 overhaul of its Listing Rules ease admission and continuing obligations,[27]

[18]Craig Doidge, G Andrew Karolyi and René M Stulz, 'The U.S. Listing Gap' (2017) 123 J Fin Econ 464; Kathleen M Kahle and René M Stulz, 'Is the US Public Corporation in Trouble?' (2017) 31 J Econ Persp 67; Mark J Roe and Charles CY Wang, 'Half the Firms, Double the Profits: Public Firms' Transformation, 1996–2022' (2025) 8 J L Fin & Acct 211; Craig Doidge and others, 'Are there too few publicly listed firms in the US?' (2025) 60 Fin Rev 317

[19]Xiaohui Gao, Jay R Ritter and Zhongyan Zhu, 'Where Have All the IPOs Gone?' (2013) 48 J Fin & Quant Anal 1663

[20]Gustavo Grullon, Yelena Larkin and Roni Michaely, 'Are US Industries Becoming More Concentrated?' (2019) 23 Rev Fin 697; Spencer Y Kwon, Yueran Ma and Kaspar Zimmermann, '100 years of rising corporate concentration' (2024) 114 Am Econ Rev 2111

[21]Jeffrey Wurgler, 'On the Economic Consequences of Index-Linked Investing' in William T Allen and others (eds), *Challenges to Business in the Twenty-First Century: The Way Forward* (American Academy of Arts and Sciences 2010); Doron Israeli, Charles MC Lee and Suhas A Sridharan, 'Is There a Dark Side to Exchange Traded Funds? An Information Perspective' (2017) 22 Rev Acct Stud 1048

[22]H Jiang, D Vayanos and L Zheng, 'Passive investing and the rise of mega-firms' (2025) 38 Rev Fin Stud 3461

[23]James A Bennett, Richard W Sias and Laura T Starks, 'Greener Pastures and the Impact of Dynamic Institutional Preferences' (2003) 16 Rev Fin Stud 1203; Anna Pavlova and Taisiya Sikorskaya, 'Benchmarking intensity' (2023) 36 Rev Fin Stud 859

[24]KJ Martijn Cremers and Antti Petajisto, 'How Active Is Your Fund Manager? A New Measure That Predicts Performance' (2009) 22 Rev Fin Stud 3329; Martijn Cremers and others, 'Indexing and active fund management: International evidence' (2016) 120 J Fin Econ 539

[25]Curtis J Milhaupt and Wolf-Georg Ringe, 'The Political Economy of Global Stock Exchange Competition' (ECGI Law Working Paper No 872/2025, 2025)

[26]Xinfei Huang, Yue Zhang and Zhe Zong, 'Governmental Venture Capital and Investor Sentiment: Evidence from Chinese Government Guidance Funds' (2025) 99 J Intl Fin Mkts Inst & Money 102120

[27]Regulation (EU) 2024/2809 of the European Parliament and of the Council of 23 October 2024 amending Regulations (EU) 2017/1129, (EU) No 596/2014 and (EU) No 600/2014 (Listing Act) [2024] OJ L 2024/2809

addressing the stock of listings rather than the flow of capital toward existing small-caps, although their durable effects remain difficult to assess at this stage.[28] Korea's Corporate Value-up Programme combines issuer disclosure, governance incentives, and index-linked infrastructure, and although it produced encouraging headline gains, the broader market effects have been uneven and concentrated among larger constituents.[29] The closest precedent for channelling public capital into listed equities is the Bank of Japan's exchange-traded-fund purchase programme, which deployed central-bank capital through a trust-bank structure into index-linked funds;[30] because that intervention operated through index replication for monetary-policy purposes rather than through active mandates aimed at an underserved segment, its benefits accrued to issuers already inside the index, so that the channel rewards issuers that can secure inclusion rather than those the policy means to reach. Read together, these programmes engage the capital channel without varying mandate specificity in a way that would isolate whether specificity governs the gap. The Singapore reform supplies that variation, since it deployed capital through active managers and then disclosed mandate terms specifying significant allocation to small- and mid-cap equities, which lets the cross-section register what changed when the terms became specific.

It is therefore surprising that the intermediation problem in public equity markets has received so little direct empirical attention, given that the managers through which such capital must pass are optimised for large, liquid, benchmark-proximate securities. Existing work treats capital allocation largely as a static outcome rather than as a quantity that responds to how the mandate is written, which leaves unexamined the design margin that governs whether policy intent survives transmission through the intermediary. Our contribution is to make that margin observable: announcement returns are consistent with an intermediation gap in a listed-equity setting, and the reversal between the February reform announcement (E1) and the July mandate disclosure (E3) indicates that the gap responds to mandate specificity.[31]

2.2 The Information Channel

A longstanding debate concerns whether securities regulation creates or destroys value, a debate that rests on the broader finding that legal institutions and investor protection shape financial development[32] and that the design of securities laws, in particular their disclosure and liability

[28] Brian R Cheffins and Bobby V Reddy, 'Law and Stock Market Development in the UK over Time: An Uneasy Match' (2023) 43 OJLS 725

[29] Bora Lee and Kian Heng Peh, 'Narrowing the Korea Discount: Stock Market Reform via Corporate Value-up' (AMRO Blog, 30 March 2026)

[30] Ben Charoenwong, Randall Morck and Yupana Wiwattanakantang, 'Bank of Japan Equity Purchases: The Final Frontier in Extreme Quantitative Easing' (NBER Working Paper No 25525, 2019); Mitsuru Katagiri, Junnosuke Shino and Koji Takahashi, 'Bank of Japan's ETF Purchase Program and Equity Risk Premium: A CAPM Interpretation' (2025) 73 J Fin Mkts 100961

[31] The full reform timeline appears in Table 2; the inferential backbone of the reversal, including the stacked interaction estimate and the paired-reversal placebo, accompanies the size exhibit in Section 5.

[32] Rafael La Porta and others, 'Legal Determinants of External Finance' (1997) 52 J Fin 1131; Rafael La Porta and others, 'Law and Finance' (1998) 106 J Pol Econ 1113; Andrei Shleifer and Daniel Wolfenzon, 'Investor Protection and Equity Markets' (2002) 66 J Fin Econ 3

provisions, governs market outcomes.[33] The two positions in that debate imply opposite predictions for how the market should price the removal of oversight. One tradition treats regulation principally as a cost borne by firms, so that deregulation should generally be welcomed: Stigler[34] and Peltzman[35] supply the theoretical foundation, and applied work bears it out, since event studies of the Sarbanes–Oxley Act document substantial compliance costs concentrated in smaller firms,[36] and the JOBS Act generated positive returns for issuers exempted from disclosure obligations.[37] Even within this account a complication is visible, because a credible commitment to stricter oversight can itself lower the cost of capital by reducing adverse selection.[38] A competing tradition treats securities regulation as informational infrastructure that lowers the cost of producing accurate prices. On this view, mandatory disclosure, fraud rules, and the monitoring instruments that accompany them help sophisticated investors and analysts maintain pricing accuracy through the arbitrage mechanisms catalogued by Gilson and Kraakman,[39] so that removing secondary-market oversight degrades that infrastructure and should be priced negatively where it applies to firms whose quality is hard to verify independently.[40] The disclosure-economics literature gives this view its empirical content, documenting that fuller disclosure reduces information asymmetry and the cost of capital,[41] that disclosure mandates generate positive abnormal returns and liquidity improvements,[42] and that enforcement intensity matters as much as the formal standard, since markets tend to price the credibility of oversight rather than its statutory text alone.[43] Symmetrically, event studies of disclosure deregulation find that withdrawing mandated reporting or oversight is priced negatively for the affected firms.[44]

[33] Rafael La Porta, Florencio Lopez-de-Silanes and Andrei Shleifer, 'What Works in Securities Laws?' (2006) 61 J Fin 1

[34] George J Stigler, 'The Theory of Economic Regulation' (1971) 2 Bell J Econ & Mgmt Sci 3

[35] Sam Peltzman, 'Toward a More General Theory of Regulation' (1976) 19 J L & Econ 211

[36] Ivy Xiying Zhang, 'Economic Consequences of the Sarbanes–Oxley Act of 2002' (2007) 44 J Acct & Econ 74; Peter Iliev, 'The Effect of SOX Section 404: Costs, Earnings Quality, and Stock Prices' (2010) 65 J Fin 1163

[37] Dhammika Dharmapala and Vikramaditya S Khanna, 'The Costs and Benefits of Mandatory Securities Regulation: Evidence from Market Reactions to the JOBS Act of 2012' (SSRN Working Paper, 2014)

[38] Cécile Carpentier, Douglas Cumming and Jean-Marc Suret, 'The Value of Capital Market Regulation: IPOs versus Reverse Mergers' (2012) 9 J Empirical Legal Stud 56

[39] Ronald J Gilson and Reinier H Kraakman, 'The Mechanisms of Market Efficiency' (1984) 70 Va L Rev 549

[40] Zohar Goshen and Gideon Parchomovsky, 'The Essential Role of Securities Regulation' (2006) 55 Duke LJ 711

[41] Paul M Healy and Krishna G Palepu, 'Information Asymmetry, Corporate Disclosure, and the Capital Markets: A Review of the Empirical Disclosure Literature' (2001) 31 J Acct & Econ 405; Christian Leuz and Robert E Verrecchia, 'The Economic Consequences of Increased Disclosure' (2000) 38 J Acct Res 91; Christian Leuz and Peter D Wysocki, 'The economics of disclosure and financial reporting regulation: Evidence and suggestions for future research' (2016) 54 J Acct Res 525; Hans B Christensen, Luzi Hail and Christian Leuz, 'Capital-Market Effects of Securities Regulation: Prior Conditions, Implementation, and Enforcement' (2016) 29 Rev Fin Stud 2885

[42] Brian J Bushee and Christian Leuz, 'Economic Consequences of SEC Disclosure Regulation: Evidence from the OTC Bulletin Board' (2005) 39 J Acct & Econ 233; Michael Greenstone, Paul Oyer and Annette Vissing-Jorgensen, 'Mandated Disclosure, Stock Returns, and the 1964 Securities Acts Amendments' (Working Paper, 2005); Allen Ferrell, 'Mandatory Disclosure and Stock Returns: Evidence from the Over-the-Counter Market' (2007) 36 J Legal Stud 213

[43] John C Coffee, 'Law and the Market: The Impact of Enforcement' (SSRN Working Paper, 2007); Hans B Christensen, Luzi Hail and Christian Leuz, 'Mandatory IFRS Reporting and Changes in Enforcement' (2013) 56 J Acct & Econ 147

[44] See Nuno Fernandes, Ugur Lel and Darius P Miller, 'Escape from New York: The Market Impact of Loosening Disclosure Requirements' (2010) 95 J Fin Econ 129; Vanessa Behrmann and others, 'The Deregulation of Quarterly Reporting and Its Effects on Information Asymmetry and Firm Value' (2025) 64 Rev Quant Fin & Acct 1221; Jörg-Markus Hitz and Florian Moritz, 'Turning Back the Clock on Disclosure Regulation? Evidence from the Termination

These competing philosophies have concrete institutional expression in how disclosure regimes assign the risk of imperfect information. When Singapore moved toward a disclosure-based approach, officials characterised it as "caveat emptor," placing the onus on investors to assess quality once issuers disclose.[45] The original United States disclosure regime framed disclosure as issuer-side verification, ensuring that "the seller also beware," which located responsibility with issuers and the institutions that bring them to market.[46] The friction evidence indicates that the market's assessment lies closer to the second view, treating the withdrawal of oversight as a shift of verification risk onto investors for firms that had attracted prior scrutiny. The two traditions are better read as jointly operative than as mutually exclusive: the same deregulation drew positive value where it removed an identified label whose institutional costs exceeded its informational content, and negative value where it removed informative monitoring.[47] The two responses are consistent only if the relevant policy variable is which mechanism is withdrawn rather than the gross quantity of oversight removed. The adverse-selection logic underlying the negative pricing is the market-for-lemons mechanism: where buyers cannot distinguish sound issuers from unsound ones, mandatory disclosure and the monitoring that supports it prevent the market from unravelling, so that withdrawing those instruments reintroduces the friction they were designed to suppress.[48]

That distinction connects the information channel to work on regulatory stigma, where the regulatory instrument itself imposes costs beyond the information it conveys. Reputational damage from regulatory sanctions can exceed the direct financial penalty, so that a visible classification may burden a firm independently of the underlying conduct that triggered it.[49] A close analogy comes from the banking literature on the discount window: institutions avoided borrowing from the Federal Reserve during the 2007–2008 crisis because the act of borrowing signalled distress, so the facility's stigma impaired the very firms it was meant to assist.[50] SGX's financial watch-list operated through a comparable mechanism, in which the label imposed financing-impairment and counterparty-reluctance costs, so that its removal could be priced as stigma relief even while the removal of genuinely informative monitoring was priced as a loss. Our findings indicate that the penalty on firms with accumulated regulatory friction is consistent, in turn, with evidence that public enforcement generates valuation spillovers through a legal-bonding channel,[51] since

of the Quarterly Reporting Mandate in Europe' (SSRN Working Paper, 2019)

[45] Lee Hsien Loong, Deputy Prime Minister, *Investor Education & New Framework for Financial Advisers*, speech at the 1st Anniversary Dinner of the Securities Investors Association (Singapore), 7 July 2000, https://www.mas.gov.sg/news/speeches/2000/investor-education-new-framework-for-financial-advisers--07-jul-2000.

[46] Franklin D. Roosevelt, Message to Congress on Federal Supervision of Investment Securities, 29 March 1933.

[47] J Lawrence, 'The Economics of Market Confidence: (Ac)Costing Securities Market Regulations' (2000) 17 Co & Sec LJ 171

[48] Akerlof (n 3); John C Coffee Jr, 'Market Failure and the Economic Case for a Mandatory Disclosure System' (1984) 70 Va L Rev 717

[49] John Armour, Colin Mayer and Andrea Polo, 'Regulatory Sanctions and Reputational Damage in Financial Markets' (SSRN Working Paper, 2010); John Armour, Colin Mayer and Andrea Polo, 'Naming and Shaming: Evidence from Event Studies' in *The Cambridge Handbook of Compliance* (Cambridge University Press 2019)

[50] Olivier Armantier and others, 'Discount Window Stigma During the 2007–2008 Financial Crisis' (2015) 118 J Fin Econ 317; Sriya Anbil, 'Managing Stigma During a Financial Crisis' (2018) 130 J Fin Econ 166

[51] Roger Silvers, 'The Valuation Impact of SEC Enforcement Actions on Nontarget Foreign Firms' (2016) 54 J Acct Res 187

withdrawing informative monitoring attenuates the bonding benefit that scrutiny had supplied. The watch-list therefore separates two functions that broad deregulation bundles together – the punitive function of a stigmatising label and the informational function of active monitoring – and the design question is whether burden removal can be made selective enough to retire the first without discarding the second, which reframes the disclosure trade-off as a question of which function to retain rather than how much oversight to keep.[52] A parallel sorting operates across listing regimes: Nielsson finds that London's AIM,[53] with its lighter listing requirements, allowed smaller firms to raise capital without attracting observably lower-quality issuers, the sorting running through firm size and sensitivity to regulatory cost, which is the same dimension along which Singapore's smaller and higher-friction firms bore the deregulation penalty. The friction results supply the primary evidence on this channel, while the watch-list evidence corroborates the distinction without serving as the principal proof.

3 Institutional Background

Singapore's 2025 equity-market reform was produced by an integrated regulatory architecture in which a single statutory authority carries a developmental mandate alongside its supervisory one. MAS is the country's integrated regulator for banking, insurance, and securities, and its enabling legislation charges it with developing Singapore as a financial centre in addition to its prudential and market-conduct functions ([MAS-STAT-CITE];[54]). Front-line responsibility for listed-market oversight is delegated to SGX RegCo, the exchange's independent regulatory subsidiary, which administers the listing rules, market surveillance, and enforcement under MAS's statutory supervision. This division of labour matters for the analysis that follows, because the reform's developmental and supervisory instruments were issued by different bodies under distinct legal authorities, a separation that Table 1 maps event by event. The institutional structure is treated here as the enabling fact that allowed capital deployment and oversight recalibration to be announced together and then separated over the following months, not as a finding.

The closest antecedent to this study is Gurrea-Martínez,[55] who provides a comprehensive qualitative assessment of the 2025 reform and situates Singapore's package within a comparative frame of market-competitiveness initiatives. That account identifies the policy instruments and their plausible effects but does not test how the market priced them. The present paper complements it by supplying announcement-return evidence on whether the reform's two channels were priced as the design intended, and by isolating the mandate-specificity margin that the qualitative literature does not measure. Singapore is well suited to this test because its concentrated ownership and

[52] Omri Ben-Shahar and Carl E Schneider, *More Than You Wanted to Know: The Failure of Mandated Disclosure* (Princeton University Press 2014) 194–95.

[53] Ulf Nielsson, 'Do less regulated markets attract lower quality firms? Evidence from the London AIM market' (SSRN Working Paper, 2013)

[54] Hans Tjio, Wai Yee Wan and Kwok Hon Yee, *Principles and Practice of Securities Regulation in Singapore* (3rd edn, LexisNexis 2017)

[55] Aurelio Gurrea-Martínez, 'Strengthening the International Competitiveness of Capital Markets: Global Insights and Local Strategies' (SMU School of Law Research Paper No 18/2025, 2025)

thin free float make intermediary investability constraints bind more tightly than in deeper markets, and because its reliance on public oversight instruments, with limited private-enforcement substitutes, gives the removal of those instruments few offsetting channels. The institutional detail that supports these features, including the watch-list mechanism and the post-listing oversight apparatus, is developed in the subsections that follow.

3.1 The Listing Drain and a History of Regulatory Tightening

SGX occupies an unusual position among Asian exchanges, combining a credible regulatory reputation with persistently weak listing activity and trading interest, even as Singapore has grown into one of the region's largest centres for private-capital formation through private equity and venture capital. That the same jurisdiction has become a preferred domicile for capital raised and deployed privately while struggling to retain and attract public listings suggests that the listing drain reflects the relative attractiveness of the public venue rather than a shortage of investable enterprise. Over the past decade, market participants and policymakers have raised recurring concerns about declining initial public offerings, thinning volumes, and market capitalisation that has stagnated relative to Hong Kong, Tokyo, and Sydney.[56] Small- and mid-cap equities bear the brunt of these self-reinforcing dynamics: thin analyst coverage limits investor awareness, the resulting low institutional participation depresses valuations and widens bid-ask spreads, and the higher transaction costs in turn deter the coverage that might have drawn institutions back, so that poor market quality both repels new listings and sustains the decline that prompted the intervention examined in this paper.[57]

The microstructure that produces these spreads is specific to a thin market, where many counters trade only intermittently and quoted depth is shallow even for firms of moderate size; Lee documents the trading processes through which SGX prices form,[58] and Claessens and others situate the development of a small exchange within the broader internationalisation of stock markets,[59] in which order flow and listings migrate toward the deepest venues. A reform that seeks to draw capital toward the lower tiers of the market therefore works against the gravitational pull of liquidity toward the largest names.

The reform must be read against a long prior period of regulatory discipline, which helps explain why the market received the 2025 package as a measured recalibration rather than a wholesale

[56] Owen Walker, 'Singapore's Stock Exchange Hits 20-Year Low in Listed Companies', *Financial Times*, 27 December 2024, https://www.ft.com/content/bd40562e-1414-485e-9a3a-b2980d7a3d4e.

[57] The structural sources of this pattern are developed in Section 2.1; the paper's own evidence on the size distribution and the analyst-coverage cliff appears in Figures 1 and 5. The two listing boards differ in admission stringency: the Mainboard imposes quantitative criteria including a minimum pre-tax-profit threshold (S$30 million, reduced to S$10 million in October 2025) and market-capitalisation requirements, while Catalist, established in 2007, operates a more flexible sponsor-supervised regime for smaller growth companies that has nonetheless attracted few high-quality listings, with many constituents trading infrequently at deep discounts.

[58] Murphy Jun Jie Lee, 'The Microstructure of Trading Processes on the Singapore Exchange' (PhD thesis, University of Technology Sydney 2013)

[59] Stijn Claessens, Daniela Klingebiel and Sergio L Schmukler, 'Stock Market Development and Internationalization: Do Economic Fundamentals Spur Both Similarly?' (2006) 13 J Empirical Fin 316

retreat. Formal securities regulation imposed minimal substantive constraints until the 1985 market crisis, when the manipulation of several large listed companies forced a three-day exchange closure; the legislative and supervisory response was a sustained tightening that continued through the 1997 Asian Financial Crisis, running counter to the broadly deregulatory trend in Western markets over the same period.[60] Singapore's corporate and securities law has historically tracked such episodes closely, a reciprocal relationship that lends reform announcements credible information content about near-term legal change.[61]

By the time the 2025 measures were announced, SGX had operated under comparatively stringent post-listing oversight for roughly seventeen years, dating from the 2008 introduction of the financial watch-list.[62] That vigilance coincided with a marked contraction in listed companies, from a peak of 782 in January 2011 to 622 by July 2024, as weaker firms exited through delistings, suspensions, and watch-list removals. The 2025 measures are therefore more accurately described as a re-liberalisation: because SGX had accumulated regulatory goodwill through this prior discipline, the market was positioned to treat the reform package as a credible adjustment of existing oversight norms rather than an abandonment of them.

3.2 Concentrated Ownership and the Investability Constraint

Concentrated ownership makes intermediary investability constraints bind more tightly in Singapore than in markets with dispersed shareholdings, and the feature connects directly to the capital channel examined below. Family control and government-linked companies are prevalent, and they leave a smaller free float available to outside institutions;[63] state ownership and concentrated blockholders interact to shape governance outcomes and the protection afforded to minority shareholders,[64] while the determinants of ownership and board structure reflect the influence of these controlling stakes on governance practice,[65] a theme pursued in subsequent work on independent directors and board composition.[66] The thin free float that concentrated ownership implies restricts the volume a fund manager can acquire without moving the price, which reinforces the tilt of intermediated capital toward larger, more widely held issuers and so tightens the investability threshold that a public programme channelled through professional managers must clear. This pattern is pronounced outside the United States, since United States index funds, given their inclusion thresholds, often hold proportionally more mid-cap than large-cap stock, whereas

[60] Tjio and others (n 54)

[61] Vincent Ooi and Cheng Han Tan, 'Singapore Company Law and the economy: reciprocal influence over 50 years' (2019) 27 Asia Pac L Rev 14

[62] Tjio and others (n 54)

[63] Cheng Han Tan, Dan W Puchniak and Umakanth Varottil, 'State-owned enterprises in Singapore: Historical insights into a potential model for reform' (2014) 28 Colum J Asian L 61

[64] Ernest WK Lim, 'Concentrated ownership, state-owned enterprises and corporate governance' (2021) 41 OJLS 663

[65] YT Mak and Yuan Li, 'Determinants of Corporate Ownership and Board Structure: Evidence from Singapore' (2001) 7 J Corp Fin 235

[66] Ernest Lim, *A case for shareholders' fiduciary duties in common law Asia* (Cambridge University Press 2019); Dan W Puchniak and Luh Luh Lan, 'Independent Directors in Singapore: Puzzling Compliance Requiring Explanation' (2017) 65 Am J Comp L 265

small-caps in Asian and European markets obtain little index exposure because of thin liquidity, transaction costs, and concentrated existing ownership.[67]

3.3 The Watch-List as a Rule-Based Classification

The financial watch-list, introduced by the exchange in 2008, is best understood as a rule-based regulatory classification with defined entry and exit criteria rather than a discretionary label. A Mainboard issuer entered the list upon satisfying two objective conditions: pre-tax losses across three or more consecutive financial years, together with an average daily market capitalisation below S$40 million over the preceding six months (an alternative twenty-cent volume-weighted-average-price criterion operated from March 2015 till June 2020).[68] Exit from the watch-list required the firm to clear the same thresholds in the other direction. The design parameters are thus the loss-duration trigger and the capitalisation floor, which jointly determined membership without case-by-case adjudication. The classification was also unusually visible by comparison with continuing-listing standards elsewhere, signalling financial distress in a market where loss-making companies are seldom funded through public equity in the first place.[69]

The classification carried substantial stigma in operation, because lenders, counterparties, and investors commonly read watch-list status as a signal of impending delisting risk, which impaired the firm's capacity to raise finance, attract investors, or restructure; within two years of the list's introduction, five of its six inaugural constituents had been delisted.[70] Membership peaked at roughly 75 firms in 2016 and declined thereafter through delistings, suspensions, and qualifying exits, leaving about 30 firms by mid-2025.[71] Because the surviving firms were themselves the residue of a protracted filtering process, the expected cost of abolition by 2025 may have been lower than at an earlier point.

The watch-list sat within a wider apparatus of post-listing instruments administered by SGX RegCo, comprising public queries issued on detection of unusual trading, trade-with-caution alerts, and trading suspensions for issuers with going-concern problems. Each instrument is individually disclosed, and together with watch-list status these interventions constitute the cumulative public record of regulatory attention that the empirical analysis treats as the firm-level measure of the

[67] Bo Becker, Rüdiger Fahlenbrach and Ehsan Mahdikhani, 'The Global Rise of Index-Based Ownership of Firms' (Working Paper).

[68] There was no equivalent watch-list for SESDAQ, later Catalist, issuers, which were instead monitored by their sponsors.

[69] Whereas Nasdaq's continuing-listing framework turns substantially on a minimum bid price, Singapore removed a comparable minimum-trading-price requirement in 2020, leaving the watch-list focused on loss-making companies with a market capitalisation below S$40 million.

[70] 'Singapore Exchange to Delist 5 Companies under Watch-List', *Singapore Business Review*, https://sbr.com.sg/financial-services/news/singapore-exchange-delist-5-companies-under-watch-list.

[71] Our equity-only regression sample contains five firms with active watch-list status at the October 2025 removal event (E4), the difference from the full list reflecting firms suspended or delisted before the event window or excluded as non-equity instruments. Only these five had at least 120 valid trading days in the estimation window for cumulative abnormal returns around E4. The five are A31, AZA, FQ7, QS9, and R14; a sixth watch-listed firm, 5GI, was delisted before the event date.

information channel. These public instruments carry particular weight in Singapore because public enforcement dominates its securities-regulation landscape, with limited private-enforcement avenues to substitute for regulatory monitoring.[72] The scarcity of private substitutes makes the abolition of a public oversight instrument a comparatively clean setting in which to observe how the market values regulatory labelling, since the removal of a public signal has fewer offsetting channels than in a jurisdiction where private litigation routinely supplements public monitoring. Recognising this scarcity, the reform package advanced investor-recourse proposals, later specified to operate through an independent designated representative coordinating action for affected investors.[73]

3.4 The Equities Market Review Group

On August 2, 2024, MAS announced the formation of the Equities Market Review Group, chaired by Mr Chee Hong Tat, then Minister for Transport, Second Minister for Finance, and Deputy Chairman of MAS, and composed of senior public- and private-sector members. The Review Group was charged with recommending measures to strengthen the competitiveness of Singapore's equities market within a twelve-month horizon, and it published its first set of measures in February 2025 and its final report in November 2025.

Two workstreams organised the Review Group's output, and each maps onto one of the reform's principal instrument types and one of the two pricing channels examined below. The Enterprise and Markets workstream addressed issuer and investor participation, liquidity, and valuation quality, pursued chiefly through capital deployment that included the S$5 billion EQDP committed by MAS and channelled to professional fund managers investing in SGX-listed equities. The Regulatory workstream addressed the listing process, the disclosure-based regime, corporate-governance standards, and investor recourse, pursued through SGX RegCo's recalibration of the post-listing oversight apparatus. The developmental instruments in the first workstream therefore rest on MAS's developmental mandate and its capital commitment ([MAS-STAT-CITE]), whereas the supervisory instruments in the second were issued by SGX RegCo under its delegated listing-regulation authority, including the prospectus lodgment and registration functions that MAS proposed to consolidate under the subsidiary while retaining statutory oversight ([MAS-STAT-CITE]). Alongside these, the package advanced listing-rule amendments reducing the Mainboard profit threshold and streamlining qualitative admission and prospectus requirements.

3.5 Timeline of Reform Events

The reform unfolded through a sequence of discrete announcements over sixteen months, set out in Table 2 and attributed to their issuing bodies in Table 1. This sequence is central to the empirical

[72] Wai Yee Wan, Christopher Chen and Say H Goo, 'Public and Private Enforcement of Corporate and Securities Laws: An Empirical Comparison of Hong Kong and Singapore' (2019) 20 EBOR 319

[73] Monetary Authority of Singapore, *Consultation Paper on Measures to Enhance Investor Recourse Avenues in Market Misconduct Cases* (24 October 2025).

design, because the first substantive announcement (E1) bundled both instrument types in a single event, while later events activated narrower and partially overlapping subsets, generating the cross-event variation in policy content that the analysis exploits.

Table 2: Timeline of Reform Events

Label	Date	Key Content	Primary Channels
E0	2 Aug 2024	Formation of Equities Market Review Group	Signal (no measures)
E1	21 Feb 2025	First measures: S$5B EQDP, disclosure-based regime shift, watch-list removal proposal, Catalist review, prospectus streamlining	Capital + Information
E2	15 May 2025	SGX RegCo consultation paper: detailed disclosure-based proposals	Information
E3	21 Jul 2025	First 3 EQDP managers named (Avanda, Fullerton, JPMAM); S$1.1B deployed; small/mid-cap mandate terms disclosed	Capital
E4	29 Oct 2025	Watch-list abolished; public trading queries ceased	Information (implementation)
E5	19 Nov 2025	Final report; SGX–Nasdaq dual listing bridge; 6 additional EQDP managers (S$2.85B); board lot reduction, market maker incentives	Capital

Notes: "Primary Channels" indicates which of the paper's two main channels (the capital channel, capital deployment through fund managers, and/or the information channel, deregulation of SGX's oversight regime) each event primarily activates. E1 bundles both; subsequent events progressively unbundle them. E5 also includes market structure enhancements (board lot reduction, market maker incentives) that operate outside the two primary channels.

The formation announcement on August 2, 2024 (E0) established the policy direction without specific measures beyond the twelve-month reporting commitment, and the paper excludes it from hypothesis testing because its window overlaps the JPY carry-trade unwind of late July and early August 2024.[74] The first substantive announcement, on February 21, 2025 (E1), set out the Review Group's measures across six pillars. On the demand side these included the S$5 billion EQDP committed by MAS, fund-manager tax exemptions, an adjustment to Option C of the Global Investor Programme, and an expansion of the GEMS research-grant scheme, alongside cross-border depository-receipt collaborations.[75] On the supply side, the regulatory pillar proposed a shift towards a disclosure-based regime through consolidation of listing review under

[74] Each event window was screened for confounding news by reviewing MAS and SGX announcements, Singapore macroeconomic releases, exchange-wide trading notices, and major regional equity-market developments within the $(-1, +3)$ window. For E1 through E5 the screen identified no concurrent Singapore or regional equity-market dislocation capable of explaining the reported cross-sectional patterns; E0 is the sole exception, for the reason given in the text.

[75] Option C of the Global Investor Programme, the family-office route, requires a Singapore-based single family office with at least S$200 million under management; from 21 February 2025 new applicants must deploy at least S$50 million into equities listed on approved Singapore exchanges. The Grant for Equity Market Singapore (GEMS) co-funds equity-research coverage of SGX-listed issuers.

SGX RegCo, streamlined qualitative admission criteria, and streamlined prospectus requirements, with the remaining pillars addressing investor recourse, shareholder engagement, and trading and settlement efficiency.

Subsequent events unbundled the February package along two parallel tracks. On the supervisory track, which activates the information channel, SGX RegCo issued its consultation paper on a more disclosure-based regime at E2 (May 15, 2025), proposing removal of the financial watch-list, a reduced Mainboard profit threshold, a shift from public queries to private engagement, time-limited trade-with-caution alerts, and narrowed suspension criteria.[76] SGX RegCo then implemented these changes with immediate effect at E4 (October 29, 2025), abolishing the watch-list and replacing public trading queries with private engagement on detection of unusual trading.[77] On the developmental track, which activates the capital channel, MAS appointed the first three EQDP asset managers (Avanda Investment Management, Fullerton Fund Management, and J.P. Morgan Asset Management) at E3 (July 21, 2025), deploying S$1.1 billion in initial capital and disclosing mandate terms specifying significant allocation to small- and mid-cap equities ([EQDP-TERMS]), while also committing S$50 million to equity research and the listed-product ecosystem and outlining proposals to enhance investor recourse.[78] At E5 (November 19, 2025) the Review Group published its final report together with the SGX–Nasdaq dual-listing bridge, which permits a primary listing on both boards using a single set of offering documents, and announced a second batch of six EQDP managers carrying S$2.85 billion alongside market-structure measures, namely a board-lot reduction and designated-market-maker incentives, that operate outside the two primary channels.[79] The dual-listing bridge raises a selection question that the present design does not resolve, because loosening access to a foreign market can attract cross-listings whose valuation consequences are mixed and partly involuntary,[80] so the bridge may sort firms in ways its designers did not intend.

4 Empirical Approach

The empirical strategy reads the reform's announcement returns as the differential cross-sectional pricing of two proxies – firm size for the capital channel and accumulated regulatory friction for the information channel – measured across the same set of SGX-listed equities at each event in

[76]Singapore Exchange Regulation, 'SGX RegCo Consults on Measures in Line with a More Disclosure-Based Regime', news release, 15 May 2025.

[77]Singapore Exchange Regulation, 'SGX RegCo Advances Disclosure-Based Regime; Proposes Rule Changes on MAS' Proposal to Consolidate Listing Review Functions under SGX RegCo', news release, 29 October 2025.

[78]Monetary Authority of Singapore, 'MAS Appoints First Batch of EQDP Asset Managers; Commits S$50 Million to Boost Equity Research and Product Listings; and Outlines Proposals to Enhance Investor Recourse', media release, 21 July 2025.

[79]Monetary Authority of Singapore, 'Review Group Completes Equities Market Review, Unveils SGX-Nasdaq Dual Listing Bridge, S$30 Million "Value Unlock" Package, and Second Batch of EQDP Asset Managers', media release, 19 November 2025.

[80]Peter Iliev, Darius P Miller and Lukas Roth, 'Uninvited U.S. Investors? Economic Consequences of Involuntary Cross-Listings' (2014) 52 J Acct Res 473

the reform timeline (Table 2).[81] What follows sets out the data and the estimation, and then states what the design can and cannot establish, with the supporting econometric detail collected in the online appendix.

Daily end-of-day prices for SGX-listed securities are drawn from SGX's proprietary data services division and span August 2022 to August 2025. Because the data licence ends in August 2025, the two events that fall after that date – the October watch-list abolition (E4) and the November final report (E5) – rely on Yahoo Finance daily closing prices, which a twenty-trading-day overlap validates against the SGX series at a median price correlation of essentially one.[82] The abnormal-return benchmark is the Fama–French three-factor series for the Asia Pacific ex Japan region,[83] taken from the Dartmouth data library; Fama and French show that regional factors capture international return premia more accurately than global factors, which supports using a regional rather than a global benchmark.[84] Firm characteristics combine SGX monthly market-capitalisation snapshots, annual financial-statement data,[85] the record of SGX regulatory actions parsed from corporate-announcement feeds, and board and sector information from SGX issuer registries.

The cross-section retains ordinary equity securities and excludes warrants, structured products, daily leverage certificates, exchange-traded funds, real estate investment trusts, and business trusts, so that it reflects the operating firms the reform was meant to reach rather than packaged or derivative instruments. This filtering leaves roughly 260 to 275 firms with valid abnormal returns at each event; a further requirement that each firm have at least 120 valid return observations in the estimation window, needed for stable factor loadings, removes the most thinly traded counters. That second requirement cuts two ways. It buys the precision in the loadings on which the abnormal returns depend, while trimming firms that concentrate in the small-cap and Catalist tiers where the channel effects are hypothesised to be strongest, so the omission makes the surviving sample conservative with respect to the documented effects.[86]

[81] The use of security-price reactions to measure the effects of regulation follows G William Schwert, 'Using Financial Data to Measure Effects of Regulation' (1981) 24 J L & Econ 121, and the event-study methodology surveyed in A Craig MacKinlay, 'Event Studies in Economics and Finance' (1997) 35 J Econ Lit 13; the same cross-sectional approach is applied to a corporate-law reform in Kenneth Khoo and Roberto Tallarita, 'The Price of Delaware Corporate Law Reform' (SSRN Working Paper No 5318203, 2025)

[82] The SGX price data, monthly listings snapshots, issuer feeds, and corporate-announcement records used here are proprietary and were obtained under a data licence from SGX covering August 2022 to August 2025. Yahoo Finance prices, used only for E4 and E5, do not carry bid–ask quotes, which precludes post-event market-quality analysis for those two events but does not affect the abnormal-return estimates. For these two events the SGX equal-weighted index used as the CAPM benchmark is likewise recomputed on the intersection of equities with valid Yahoo observations ($\approx$716 stock codes), so the composition shift affects only the CAPM market return there; the externally sourced Fama–French factors are unaffected.

[83] Eugene F Fama and Kenneth R French, 'Common Risk Factors in the Returns on Stocks and Bonds' (1993) 33 J Fin Econ 3

[84] Eugene F Fama and Kenneth R French, 'Size, value, and momentum in international stock returns' (2012) 105 J Fin Econ 457

[85] Annual financial-statement data are drawn primarily from Worldscope via WRDS and supplemented by figures extracted from approximately 17,000 SGX filing PDFs through a three-pass pipeline: `pdfplumber` for structured tables (roughly a 70% success rate), a Qwen2.5-VL vision model run via local Ollama inference for unstructured layouts, and manual review flags for approximately 70 rows with known scale or currency ambiguities.

[86] Among firms excluded for insufficient trading, the median count of valid return days is fewer than sixty against a calendar of roughly 280 trading days. The derivation of the threshold and the precision–coverage trade-off it reflects

The two channels enter the cross-section through firm-level proxies. The capital channel is proxied by firm size, measured primarily by an above-median indicator – equal to one when a firm's log market capitalisation exceeds the event-specific sample median – with continuous log market capitalisation retained as a secondary measure, both taken from the most recent monthly snapshot before the event window. The information channel is proxied by a regulatory friction count, that is, the cumulative number of SGX queries, trading halts, and trading suspensions a firm had attracted as of the event date, which two later decompositions sharpen by separating routine queries from the more consequential halts and suspensions, and by separating friction accumulated in the preceding twelve months from older friction.[87] A watch-list indicator marks the five firms carrying SGX's Financial Watch-List status at the October abolition event, and the regressions additionally condition on standardised illiquidity,[88] a Catalist listing indicator,[89] accounting controls, and two-digit-industry sector fixed effects.

For each event, firm-level cumulative abnormal returns are the residuals from the regional three-factor benchmark, adjusted for thin trading and cumulated over a five-day window running from one trading day before to three trading days after the announcement.[90] Writing the thin-trading-adjusted firm and factor returns with an asterisk and the corresponding intercept regressor as w_t, the benchmark is

$$R^*_{i,t} = \hat{\alpha}_i \, w_t + \hat{\beta}_{1,i} \, \mathrm{MKT}^*_t + \hat{\beta}_{2,i} \, \mathrm{SMB}^*_t + \hat{\beta}_{3,i} \, \mathrm{HML}^*_t + \hat{\varepsilon}_{i,t}, \tag{1}$$

and the residuals cumulated over the event window are the firm's abnormal return. The thin-trading correction follows Maynes and Rumsey[91] and addresses the non-synchronous trading common on SGX, where many securities do not trade every day.[92] The fixed pre-E1 estimation window that protects the factor loadings from contamination by the reform itself is detailed in the online appendix. A single-factor market model benchmarked to an SGX equal-weighted index serves as a secondary specification in the watch-list analysis: there the three-factor model's size factor mechanically absorbs the variation of interest, because the watch-list eligibility criterion is

are set out in the online appendix.

87 The friction count is built from eight SGX "Announcement Category" classes, matched exactly on the structured CSV feed: (1) Response to SGX Queries, (2) Query Regarding Trading Activity, (3) Request for Trading Halt, (4) Request for Lifting of Trading Halt, (5) Request for Suspension, (6) Request for Resumption of Trading from Suspension, (7) Regulatory Actions by SGX, and (8) Announcement in Relation to Regulatory Actions by SGX and/or Other Authorities. Friction accumulates over August 2022 to August 2025 for E0–E3 and is extended through the event date for E4 and E5. The type decomposition groups (1)–(2) as queries, (3)–(6) as halts and suspensions, and (7)–(8) as regulatory actions.

88 Yakov Amihud, 'Illiquidity and stock returns: cross-section and time-series effects' (2002) 5 J Fin Mkts 31

89 The Catalist board-tier effect is an artefact of sample composition and is absorbed by firm size; see Table 24.

90 Estimates under alternative event windows are reported in Table 19.

91 Elizabeth Maynes and John Rumsey, 'Conducting event studies with thinly traded stocks' (1993) 17 J Banking & Fin 145

92 When a firm does not trade for d_t consecutive days, the procedure cumulates the firm's return and the factor returns over the non-trading gap and divides all variables by $\sqrt{d_t}$ to correct for the heteroskedasticity induced by the unequal return horizons. It omits the conventional constant and instead includes a thin-trading-adjusted intercept regressor, $w_t = 1/\sqrt{d_t}$, because in a thin market a conventional intercept can absorb genuine abnormal returns into the alpha estimate when many zero-return days are present.

itself defined partly by market capitalisation.[93] Descriptive statistics for the cross-section, and the firm-size distribution that motivates the median split, are reported in Table 3 and Figure 1. The cross-sectional regressions then relate these abnormal returns to the size and friction proxies through a common set of specifications, with the deeper estimation detail – including the concern that the abnormal returns enter as estimated quantities[94] and the bootstrap that addresses it – collected in the online appendix.

The regressions build up in steps across the main tables, so that the stability of each estimate can be followed as controls are added. First, the variable of interest enters on its own to establish the unconditional association; second, the other channel's variable is added to test whether the two absorb one another; and third, further controls follow for firm financials, market microstructure and listing tier, and industry sector. Because the two channel variables enter together once both are present, the progression doubles as a test of whether capital and information are separable pricing dimensions or two labels for one underlying characteristic. Where adjacent events are pooled, as in the February-to-July reversal, the estimates allow for each firm contributing a correlated observation at both events. Abnormal returns are trimmed at the extreme percentiles to limit the influence of outliers, robust standard errors are used throughout, and inference on the headline coefficients is corroborated by a bootstrap that carries the first-stage estimation error into the second stage.

Table 3: Descriptive Statistics (E1 Sample)

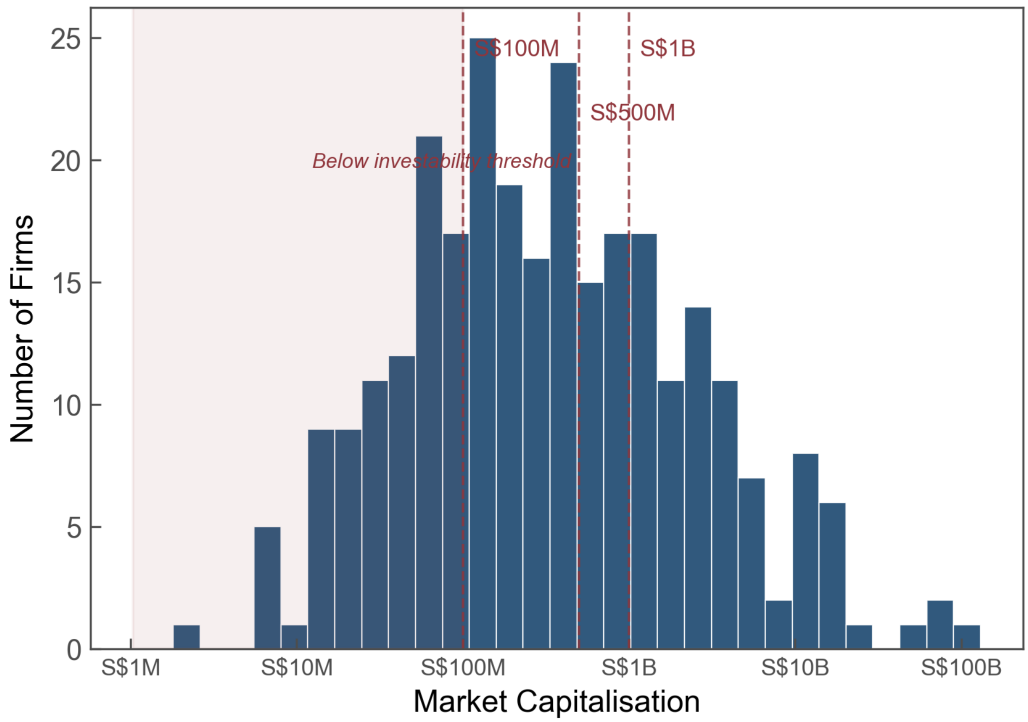
	N	Mean	SD	Min	Median	Max
CAR($-1, +3$)	274	-0.013	0.055	-0.195	-0.006	0.115
log(Market cap)	283	19.637	1.958	15.827	19.501	24.774
Friction count	285	2.475	3.302	0.000	1.000	17.160
Watchlist	285	0.018	0.132	0.000	0.000	1.000
Above median	285	0.498	0.501	0.000	0.000	1.000
ROA (%)	257	1.848	14.242	-78.143	3.855	19.459
Leverage	258	23.874	18.016	0.000	22.869	67.866
Book-to-market	249	1.540	1.111	0.032	1.325	5.034
Illiquidity (z-Amihud)	283	-0.176	0.159	-0.217	-0.217	0.749
Catalist	285	0.147	0.355	0.000	0.000	1.000

Notes: Sample comprises 285 SGX-listed equities in the E1 cross-section (2025-02-21); 274 have valid FF3 CARs. Market cap is from the last available trading day before the event window. Illiquidity (z-Amihud) is the standardised Amihud illiquidity measure over the 60 trading days before the event. Friction count is the cumulative number of SGX queries, trading halts, and trading suspensions accumulated by the firm as of the event date; watch-list is reported as an indicator variable. Broker ownership is the share of top-20 holdings registered through retail securities brokers; available for 205 firms with shareholder register data. Accounting variables (ROA, Leverage) are available for firms with Worldscope coverage ($N = 257$–258). Catalist and watch-list are indicator variables.

[93] The core size and friction results are robust across the alternative factor models reported in Table 20.

[94] Adrian Pagan, 'Econometric Issues in the Analysis of Regressions with Generated Regressors' (1984) 25 Intl Econ Rev 221

Figure 1: Distribution of market capitalisation across the SGX equity-only sample



Note: The figure illustrates the distribution of market capitalisation across the SGX equity-only sample, shown on a log-scaled x-axis. Market capitalisation is from the last available SGX monthly snapshot before E1. Dashed vertical lines mark illustrative investability thresholds at S$100 million, S$500 million, and S$1 billion; the shaded region highlights firms below the S$100 million threshold.

The evidence should be read as the differential cross-sectional pricing of the capital and information proxies, subject to clear limits, rather than as a measurement of the value of each policy function. The design rests primarily on cross-sectional decomposition within the bundled February announcement (E1), where size and friction load at the same event. The intention to separate the two channels across later events is only partly realised. The May consultation (E2) produced a null average response once detailed proposals had been signalled; the July manager appointment (E3) reversed the size gradient rather than isolating the capital-deployment channel cleanly; and the November event (E5) bundles market-structure measures with further capital deployment and the dual-listing bridge in a way that prevents clean attribution of those measures. The investability characteristics that the capital channel is meant to capture – size, analyst coverage, and foreign institutional ownership – are strongly collinear. In this context, we treat investability as a single composite proxied by firm size and use coverage and ownership only as corroborating evidence. The watch-list result rests on five treated firms, and is therefore suggestive of stigma relief rather than precisely estimated. All findings are specific to the SGX setting and need not extend to markets with different investor bases, regulatory architectures, or levels of secondary-market liquidity.

5 Findings

The findings are organised around the two channels and their behaviour across the reform sequence. After summarising the average market response at each event, we trace the intermediation gap and its reversal once the mandate terms were disclosed, the information-loss pricing of deregulation, and the watch-list exception that locates the design margin on the information channel. Throughout, the friction results carry the analytical weight on the information channel, and the watch-list evidence is treated as corroborative given its small treated sample.

Table 4 reports the average abnormal return at each event. The bundled February announcement (E1) produced a mean cumulative abnormal return of −1.32 percent, significant at the 1% level, with only 40.5% of firms posting positive returns: the typical firm lost value on the news. The May consultation (E2) drew a null response (−0.11 percent, $p = 0.721$), consistent with its having been anticipated once E1 signalled that detailed proposals would follow. The July manager appointment (E3) produced a positive mean of 3.12 percent with 70.0% of firms gaining, the October watch-list abolition (E4) a positive 1.20 percent, and the November final report (E5) a positive 0.62 percent, each significant at the 1% level. The movement from a negative average at E1 to a positive average at E3 is what motivates the cross-sectional analysis that follows, because the aggregate figures conceal the size- and friction-based differentiation the channels predict.[95]

[95] Event-level cross-sectional detail for E0, E2 and E3, and E5 is reported in Sections A.4.1 to A.4.3.

Table 4: Event-Level Summary Statistics

Event	Date	Channel	N	Mean CAR	t-stat	p-value	Median	% Pos
E0	2024-08-02	Signal	260	-0.0215^{***} (0.0041)	-5.25	0.000	-0.0187	30.8%
E1	2025-02-21	Capital + Information	274	-0.0132^{***} (0.0033)	-3.95	0.000	-0.0058	40.5%
E2	2025-05-15	Information	275	-0.0011 (0.0031)	-0.36	0.721	-0.0037	42.9%
E3	2025-07-21	Capital	270	0.0312^{***} (0.0046)	6.82	0.000	0.0199	70.0%
E4	2025-10-29	Information (implementation)	258	0.0120^{***} (0.0040)	2.99	0.003	0.0035	55.8%
E5	2025-11-19	Capital	258	0.0062^{***} (0.0022)	2.76	0.006	0.0063	60.5%

Notes: The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses; significance is assessed by the Student t-test. % Pos is the percentage of firms with positive CARs at that event. "Channel" indicates which of the paper's two main channels—the capital channel (deployment through fund managers) and the information channel (deregulation of SGX's oversight regime)—each event primarily activates. E5 also includes market structure enhancements (board lot reduction, market maker incentives) that operate outside the two primary channels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ for H_0: mean = 0.

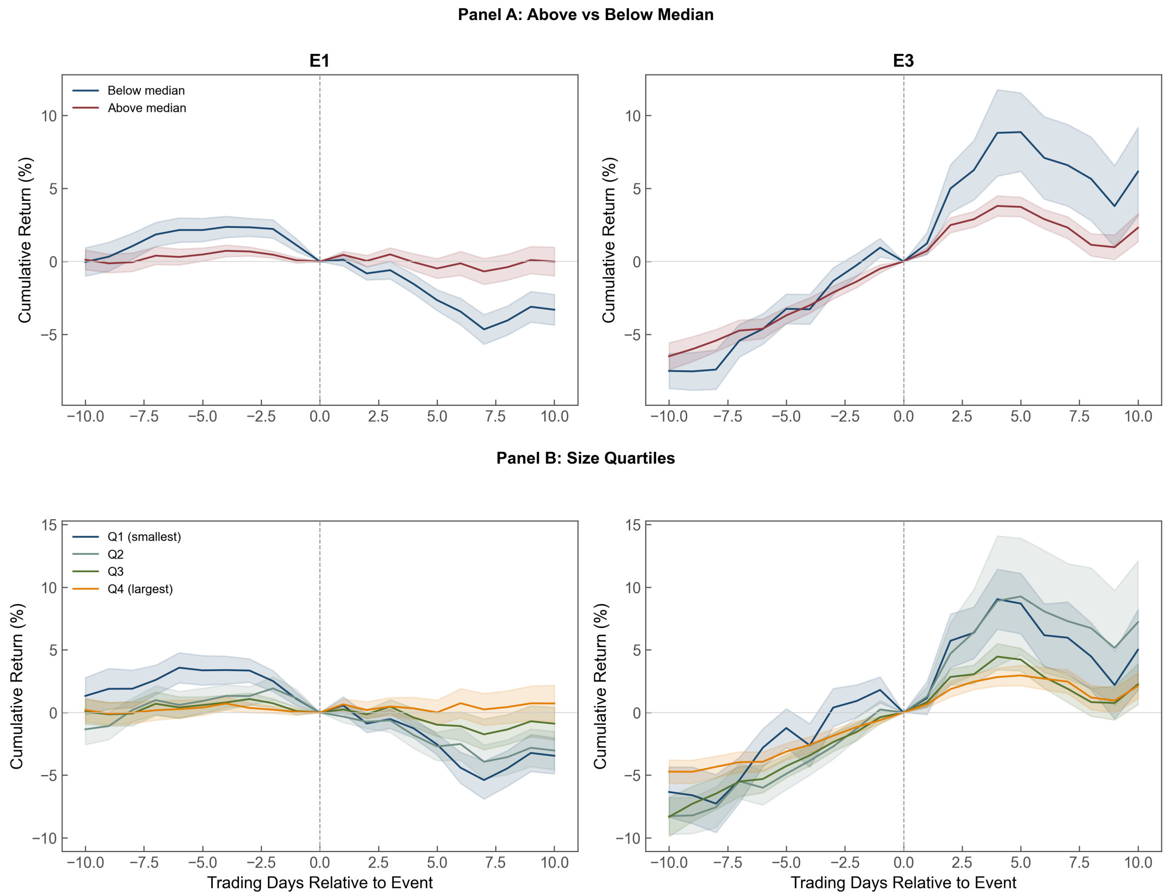
5.1 The Intermediation Gap and the Mandate-Specificity Reversal

At announcement, the capital programme was priced as a large-cap event. The EQDP committed S$5 billion through professional managers operating under standard institutional constraints – fiduciary obligations that favour established firms on a risk-adjusted basis,[96] fee structures tied to assets under management that penalise small positions, and liquidity screens that exclude thinly traded securities from the investable universe. Had the market anticipated these constraints, the announcement should have produced a cross-sectional pattern in which larger firms gained disproportionately, irrespective of the stated objective of supporting small- and mid-cap equities. The data display that pattern directly. Sorting firms into quartiles of market capitalisation within the E1 cross-section, the smallest quartile earned a mean cumulative abnormal return of −2.54 percent, significant at the 1% level, while the largest quartile gained 0.22 percent, which is statistically indistinguishable from zero, so the spread between the extreme quartiles was 2.76 percentage points, significant at the 1% level (Figure 3). The gradient is monotonic across the four quartiles rather than confined to the tails, which is what a binding investability constraint – as opposed to an idiosyncratic reaction of the very smallest firms – would generate. The quartile means are reported in Table 5, and the daily cumulative abnormal-return paths in Figure 2 trace the E1

[96] Law Commission, *Fiduciary Duties of Investment Intermediaries* (Law Com No 350, 2014)

decline and the E3 reversal directly.

Figure 2: Cumulative abnormal return paths for above- and below-median firms across E1 and E3



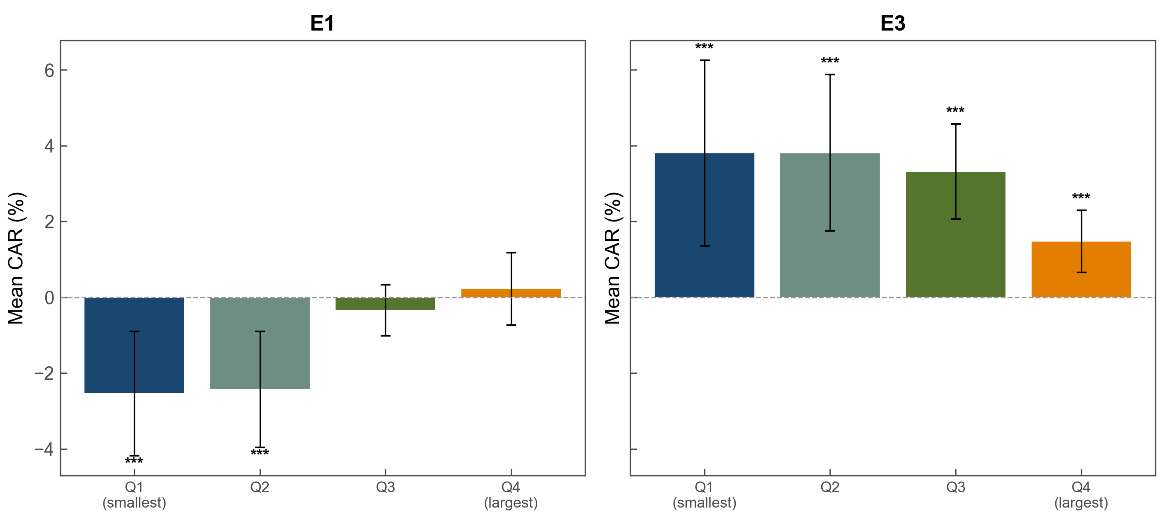
Note: Panel A: Daily cumulative log returns for firms above (navy) vs. below (maroon) median log market capitalisation, normalised to zero at day 0. Panel B: Daily cumulative log returns by size quartile (Q1 = smallest). Left column: E1 (February 21, 2025). Right column: E3 (July 21, 2025). The E1-to-E3 reversal in the size gradient is visible in both panels.

Table 5: Size Quartile CARs Across E1 and E3

Quartile	E1	E3
Q1 (smallest)	-2.54^{***}	$+3.83^{***}$
	(0.85)	(1.27)
Q2	-2.43^{***}	$+3.83^{***}$
	(0.79)	(1.07)
Q3	-0.34	$+3.32^{***}$
	(0.34)	(0.64)
Q4 (largest)	$+0.22$	$+1.48^{***}$
	(0.49)	(0.42)
$Q4 - Q1$	$+2.76^{***}$	-2.36^{*}
	(2.83)	(-1.76)
N	272	269

Notes: Mean cumulative abnormal returns (%) estimated over the $(-1, +3)$ trading-day window around the event date using a Fama–French three-factor model with Asia Pacific ex Japan factors. Size quartiles formed on log market capitalisation within each event. Conventional standard errors of the mean $(s/\sqrt{n})$ are reported in parentheses for quartile means; significance is assessed by the Student t-test. Welch t-statistic in parentheses for the $Q4 - Q1$ difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 3: Mean cumulative abnormal returns by market capitalisation quartile across E1 and E3



Note: Mean FF3 CAR$(-1, +3)$ by market capitalisation quartile. Error bars show 90% confidence intervals. At E1, returns increase monotonically with size ($Q4 - Q1 = 2.76$ pp, $p < 0.01$). At E3, the gradient reverses, with smaller firms gaining more ($Q1 = 3.83\%$, $Q4 = 1.48\%$), consistent with the mandate terms targeting small- and mid-cap allocation. The stacked E1–E3 interaction estimates a change in the above-median effect of $\gamma_2 = -0.042$, significant at the 1% level (online appendix, Table 8), and a paired-date placebo places the observed reversal at the 2nd percentile of its null distribution ($p = 0.030$; online appendix, Table 16).

The cross-sectional regression confirms the sort. Comparing means across the median partition, firms above the median of log market capitalisation experienced mean abnormal returns of −0.08 percent at E1 while those below experienced −2.57 percent, a statistically significant spread of 2.49 percentage points (Table 7). In the regression itself, the above-median size indicator carries a coefficient of roughly 2.3 percentage points, significant at the 1% level, which remains stable as financial controls and the friction and watch-list variables are added in turn (Table 6). The continuous size measure tells the same story. A threshold rather than a log-linear specification best captures the above-median pattern, because the gradient is steep below the median and flat above it; the continuous coefficient is 0.0054 per log-unit, significant at the 1% level (Tables 21 and 22). This configuration matches the prediction that capital routed through professional managers would favour firms already inside the investable universe: the managers' fiduciary obligations, fee structures, liquidity screens, and benchmark constraints together make small-cap deployment costly relative to its contribution to a benchmark-referenced portfolio.

Table 6: Size and Announcement Returns: Regression Specifications (E1 and E3)

	E1					E3				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Above median	0.0249***	0.0227***	0.0211***	0.0175***	0.0203**	-0.0175*	-0.0190**	-0.0148	-0.0127	0.0025
	(0.0065)	(0.0062)	(0.0071)	(0.0067)	(0.0096)	(0.0098)	(0.0094)	(0.0102)	(0.0101)	(0.0117)
Friction count		-0.0031**	-0.0034**	-0.0031**	-0.0032**		0.0004	0.0013	0.0009	0.0015
		(0.0012)	(0.0016)	(0.0012)	(0.0016)		(0.0016)	(0.0019)	(0.0016)	(0.0015)
Watchlist		0.0524*	0.0575	0.0512	0.0559		-0.0612	-0.0763	-0.0126	0.0100
		(0.0316)	(0.0360)	(0.0327)	(0.0370)		(0.0697)	(0.0744)	(0.0692)	(0.0655)
Illiquidity				-0.0018	-0.0014				-0.0203*	-0.0253***
				(0.0098)	(0.0110)				(0.0106)	(0.0090)
Catalist				-0.0098	-0.0001				0.0199	0.0200
				(0.0126)	(0.0176)				(0.0192)	(0.0245)
Financial controls	No	No	Yes	No	No	No	No	Yes	No	No
Sector FE	No	No	No	No	Yes	No	No	No	No	Yes
N	274	274	240	272	243	270	270	238	268	248
R^2	0.051	0.099	0.135	0.099	0.205	0.013	0.023	0.051	0.043	0.246

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around the event date, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Columns (1)–(5): E1; Columns (6)–(10): E3. Above median = 1 if $\ln(\text{MktCap})_i \geq$ sample median within each event. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Mean CARs by Size Group (E1 and E3)

	E1	E3
Below median	-2.57^{***}	$+4.10^{***}$
	(0.58)	(0.88)
Above median	-0.08	$+2.34^{***}$
	(0.29)	(0.42)
Difference	$+2.49^{***}$	-1.75^{*}
	(3.82)	(-1.79)
N	274	270

Notes: Mean cumulative abnormal returns (%) by above/below median log market capitalisation, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors over the $(-1, +3)$ trading-day window. Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses for group means; significance is assessed by the Student t-test. Welch t-statistic in parentheses for the two-sample difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The gradient reversed once the mandate terms became specific. At the July appointment of the first three EQDP managers, whose mandate terms specified significant allocation to small- and mid-cap equities, the smallest quartile gained 3.83 percent and the largest 1.48 percent, both significant at the 1% level, so the size spread turned negative and marginally significant at −2.36 percentage points (Figure 3). The regression mirrors the sort: the binary split that favoured above-median firms by 2.49 percentage points at E1 reverses to a below-median advantage of 1.75 percentage points at E3, the above-median coefficient reaches −1.9 percentage points at the 5% level in the E3 cross-section, and the friction count carries no explanatory power at E3 (Table 6). Pooling E1 and E3 in a stacked design in which each firm is observed at both events, the above-median indicator loads at $\gamma_1 = 0.025$ at E1 and its E3 interaction at $\gamma_2 = -0.042$, both significant at the 1% level, so the size advantage of 2.5 percentage points at announcement becomes a disadvantage of 1.7 percentage points at implementation. This interaction is stable as controls accumulate, ranging only from −0.047 to −0.038 as financial controls, the friction count, and sector effects are added in turn, all significant at the 1% level (Table 8). The estimate is therefore not an artefact of any particular control set, and it coincides quantitatively with the placebo reversal statistic discussed next.

Table 8: Stacked E1–E3 Size Interaction

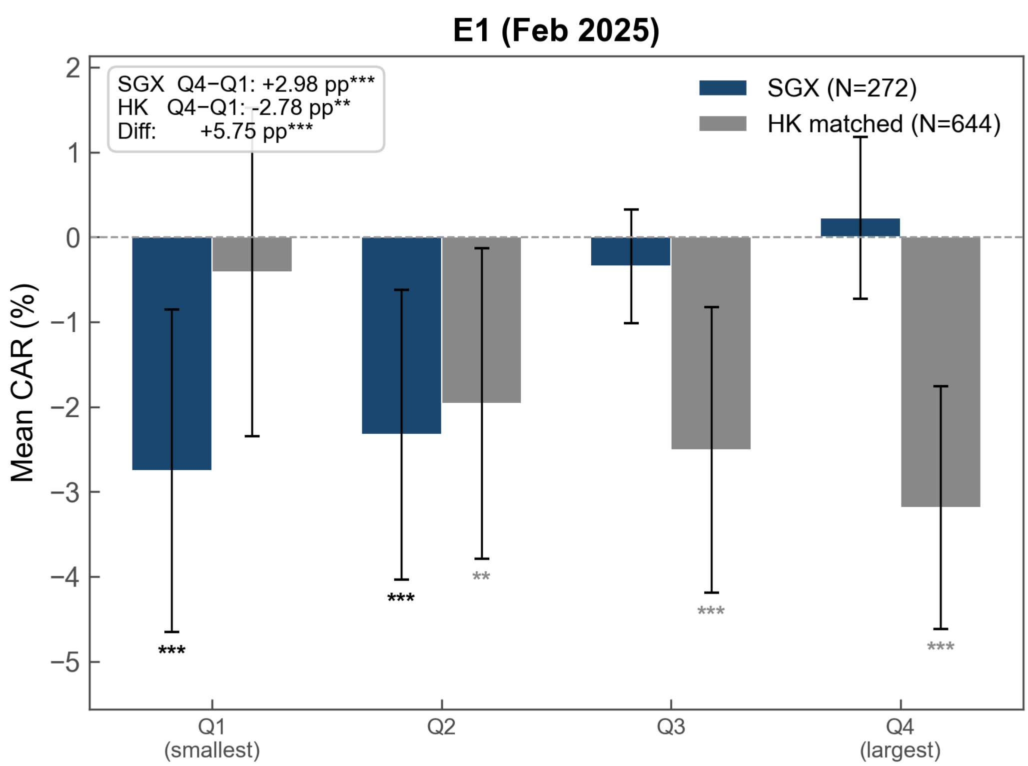
	(1)	(2)	(3)	(4)
Above median	0.0249***	0.0280***	0.0232***	0.0264***
	(0.0065)	(0.0078)	(0.0071)	(0.0092)
Post-E3	0.0667***	0.0699***	0.0557***	0.0542***
	(0.0105)	(0.0111)	(0.0112)	(0.0114)
Above median × Post-E3	-0.0424***	-0.0467***	-0.0397***	-0.0377***
	(0.0116)	(0.0122)	(0.0110)	(0.0113)
Friction count			-0.0033**	-0.0036*
			(0.0016)	(0.0021)
Friction × Post-E3			0.0044	0.0045
			(0.0032)	(0.0035)
Financial controls	No	Yes	Yes	Yes
Sector FE	No	No	No	Yes
N	544	478	478	471
R^2	0.126	0.142	0.155	0.249

Notes: OLS estimates of the stacked E1–E3 size specification with event fixed effects (captured by Post-E3 = 1 for E3 observations) and standard errors clustered by stock code, because each firm contributes two observations (one at E1, one at E3); HC1 would understate the standard errors by treating these as independent, when in reality the same firm's CARs at E1 and E3 are correlated. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Column (1) reports the bare stacked specification. Column (2) adds financial controls (ROA, leverage, and book-to-market). Column (3) further adds the friction count and its E3 interaction as a robustness check; this column corresponds to column (3) of Table 13. Column (4) augments column (3) with two-digit SSIC sector fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The reversal is the paper's central empirical hinge. Two placebo comparisons place it outside the range of variation that chance cross-sectional movement would produce. A paired-date placebo samples two hundred pairs of pseudo-event dates from the pre-reform period, each pair separated by the E1–E3 interval of roughly 105 trading days, estimates the same cross-sectional specification at both dates of each pair, and records the change in the size coefficient. The observed above-median reversal of $\Delta\hat{\beta} = -0.042$ falls at the second percentile of this null distribution ($p = 0.030$), with only six of the two hundred placebo pairs generating a reversal of comparable magnitude, and the continuous-size reversal of $\Delta\hat{\beta} = -0.011$ is equally unusual ($p = 0.030$) (online appendix, Table 16). The placebo also rules out two competing readings of the sequence. The first holds that small-cap SGX equities were oversold before E3 and mean-reverted on positive news rather than repricing mandate credibility; because the paired-date design subsumes any such tendency for small firms to bounce after declines into the null distribution, the observed reversal still sits in the tail against it. The second attributes the pattern to a generic small-cap rebound unrelated to the mandate, yet the event sequence is difficult to reconcile with that account, since returns to small firms were negative at E1, null at E2, and positive at E3 precisely when the mandate terms were disclosed, whereas a generic rebound would not be timed to the disclosure.

The remaining market-wide alternative is a regional size rotation rather than a Singapore-specific treatment effect, and a cross-market placebo addresses it. Matching each SGX firm to its three nearest Hong Kong Exchange neighbours by log market capitalisation produces a matched sample of 644 Hong Kong firms, and the identical size–CAR specification estimated on both markets at E1 yields opposite gradients (Figure 4). Whereas the SGX sample shows the monotonic upward gradient documented above, the matched Hong Kong sample shows a significantly negative size gradient, with an extreme-quartile spread of -2.78 percentage points ($p = 0.024$) – opposite in sign to the SGX pattern – and the cross-market difference in the above-median coefficient is significant at the 1% level (online appendix, Table 17). A common regional shock to small-cap valuations cannot generate gradients of opposite sign in two neighbouring markets matched on size, so a regional rotation does not account for the SGX result.

Figure 4: Cross-Market Placebo: Size Quartile CARs at E1 (SGX vs Matched HK)



Note: Bars show mean FF3 CARs (%) by size quartile at E1 (February 2025). Navy bars: SGX firms. Gray bars: HKEX firms matched by log market capitalisation ($K = 3$). Full regression results are available in Table 17.

Although they do not enter the regression jointly, two further dimensions of investability move with size in the same way and corroborate the interpretation. Sorting firms by analyst coverage, those with no coverage – and therefore invisible to the institutional research ecosystem – earned mean abnormal returns of −1.80 percent at E1, significant at the 1% level, while firms followed by six or more analysts earned −0.42 percent, which is statistically indistinguishable from zero; at E3, where the GEMS research-grant expansion specifically targeted under-covered firms, the gradient reversed, with zero-coverage firms gaining 3.04 percent against 1.05 percent for the most covered (Spearman $\rho = -0.182$, $p = 0.060$) (Table 9). Foreign institutional ownership traces the same E1 gradient, with top-quartile firms gaining 0.20 percent and bottom-quartile firms losing 2.36 percent (Spearman $\rho = 0.129$, $p = 0.083$). Because each measure is strongly collinear with size, neither coverage nor foreign ownership retains significance once log market capitalisation is controlled, which is why the paper treats investability as a single composite proxied by firm size and uses coverage and ownership only as corroborating evidence. The pattern accords with the mechanism identified by Didier,[97] who shows that foreign institutional investors tend to concentrate in large firms because size proxies for the information environment that enables institutional participation, so the three proxies capture one underlying construct rather than three separable channels.[98]

[97]Tatiana Didier, 'Information Asymmetries and Institutional Investor Mandates' (Working Paper, 2011)

[98]Retail-dominated firms, proxied by high broker ownership, display the same E1 discount and E1-to-E3 reversal; see Tables 26 to 28.

Table 9: Analyst Coverage and Foreign Institutional Ownership

Panel A: Mean CARs (%) by I/B/E/S Analyst Coverage Group

	E1 (Feb 2025)		E3 (Jul 2025)	
Coverage group	Mean CAR	N	Mean CAR	N
Zero coverage	-1.80^{***}	165	$+3.04^{***}$	163
	(0.48)		(0.58)	
1–2 analysts	-0.65	57	$+2.81^{***}$	51
	(0.81)		(0.69)	
3–5 analysts	-0.23	21	$+3.50^{**}$	22
	(0.75)		(1.64)	
6+ analysts	-0.42	31	$+1.05^{**}$	34
	(0.49)		(0.43)	
Spearman ρ (covered only)	-0.083	109	-0.182^{*}	107
	(0.096)		(0.096)	

Panel B: Foreign Institutional Ownership Quartile CARs (E1 only)

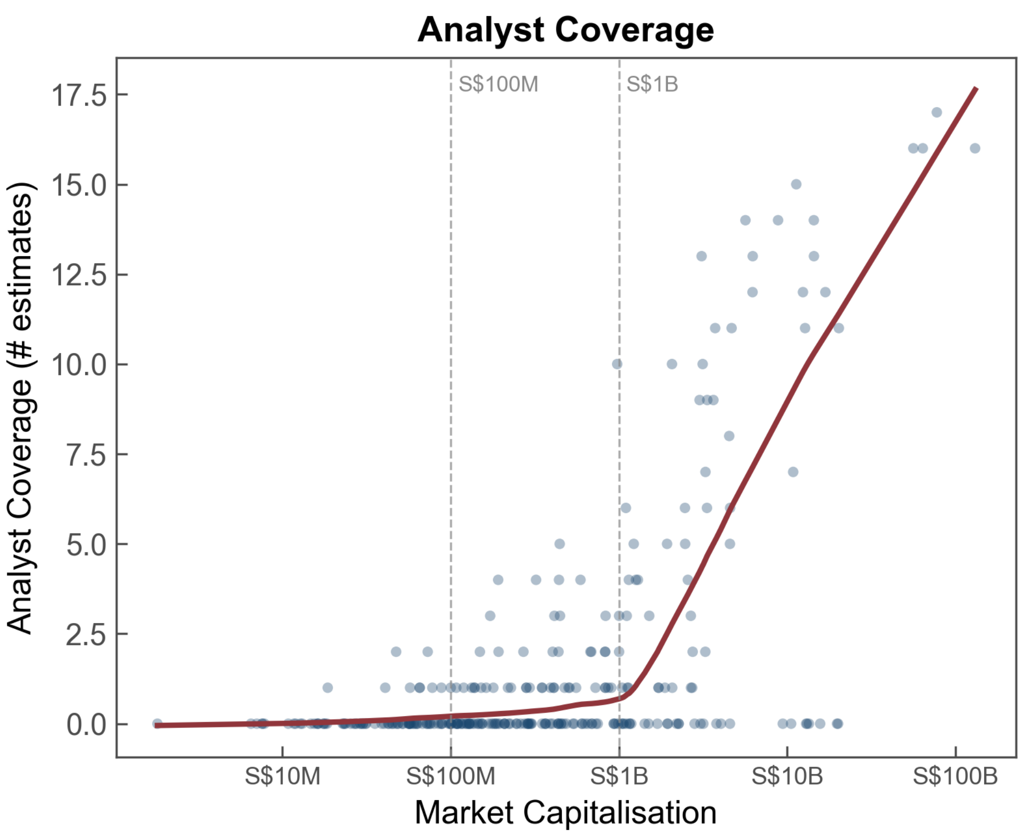
	Mean CAR	Mean IO_{for}	N
Q1 (lowest)	-2.36^{**}	0.01%	46
	(0.96)		
Q2	-0.67	0.23%	44
	(1.02)		
Q3	-0.85	1.35%	45
	(0.70)		
Q4 (highest)	$+0.20$	9.72%	46
	(0.68)		
Spearman ρ	$+0.129^{*}$		181
	(0.074)		

Notes: Panel A: Mean CARs (%) by I/B/E/S analyst coverage group. Firms not in I/B/E/S are classified as zero coverage. Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses for group means; significance is assessed by the Student t-test. Asymptotic standard errors in parentheses for Spearman ρ (computed on covered firms only). Panel B: E1 only. Foreign institutional ownership (IO_{for}) from FactSet, most recent quarter $\leq$ event date (Q4 2024). Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses for quartile means; asymptotic standard error in parentheses for Spearman ρ. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The size dispersion of the SGX universe explains why the constraint operates as a threshold. With a total market capitalisation of just under S$1 trillion, the universe ranges from micro-cap firms valued at under S$2 million to constituents exceeding S$130 billion, and in a market with this degree of dispersion firm size largely determines institutional visibility and tradability. Firms below roughly S$100 million in capitalisation typically attract little institutional visibility or tradability, whereas firms above S$1 billion generally clear those constraints; correspondingly, sell-side coverage is near zero below approximately S$1 billion before rising sharply above it (Figure 5).

The gradient between these two states is steep enough to support a threshold interpretation rather than a continuous one, which is why the binary above-median indicator outperforms the continuous measure in precision. The contrast with a deep market is instructive. In the United States, public capital channelled through fund managers can reach mid-sized firms, because those firms already sit above the visibility threshold and the manager's task is to overweight them; in Singapore, by contrast, the intended beneficiaries sit below the threshold and outside the investable universe. The intermediation gap therefore follows directly from the architecture of professional asset management, whose compliance screens, liquidity requirements, and benchmark constraints render small SGX-listed firms inaccessible irrespective of the policy mandate.

Figure 5: Analyst Coverage by Market Capitalisation



Note: Each point represents one SGX-listed firm. The I/B/E/S analyst estimate count rises sharply above approximately S$1 billion in market capitalisation. Dashed lines at S$100 million and S$1 billion mark approximate thresholds.

The reversal narrowed the gap without closing it. Cumulated across the February announcement and the July disclosure, the net size effect still favoured above-median firms by roughly 0.7 percentage points, so mandate specificity offset the intermediation bias rather than eliminating it.[99] The announcement reversal, its stability across specifications, and the two placebo comparisons

[99]A descriptive analysis of realised dollar volume across the reform sub-periods, reported in Table 29 and Figure 9 (see Section A.4.6), shows below-median firms gaining volume fastest during the active-deployment window.

together indicate that the market repriced the same programme once its terms credibly constrained the intermediaries. This is the intermediation gap in announcement-return form: the market first priced the intermediary constraints, then repriced the same commitment when the mandate disclosed terms that bound those constraints.

5.2 Deregulation as Information Loss

The deregulatory component engages two competing views of what regulatory oversight does. On the view that friction is principally a cost imposed on firms, reducing it should benefit the most heavily monitored firms, so that high-friction firms gain when oversight is withdrawn. On the competing view that the public oversight apparatus produces information the market uses to price firms whose quality is hard to verify, withdrawing it removes a monitoring signal, so that high-friction firms are penalised when removal is proposed. The second prediction has an adverse-selection structure: a public query tells the market which firms the regulator has flagged, and once the query regime is removed investors can no longer separate clean firms from problematic ones that go unflagged. The result depresses valuations across the previously monitored segment and recreates the market-for-lemons problem that disclosure and monitoring regimes exist to suppress.[100]

The cross-section supports the information reading. Table 10 reports the regression of E1 abnormal returns on the regulatory friction count alongside the size and watch-list variables.[101] Each additional friction event predicts a decline of roughly one-third of a percentage point, significant at the 1% level in the univariate specification, and the coefficient is stable between −0.0031 and −0.0034 as the size indicator, the watch-list indicator, financial controls, illiquidity, the Catalist indicator, and sector fixed effects are added. The size indicator and the friction count load simultaneously, and neither absorbs the other, so the reform was priced on two statistically independent dimensions – capital through size and information through friction.[102] Because the watch-list eligibility criterion is defined partly by market capitalisation, the three-factor model's size factor mechanically absorbs part of the watch-list variation, which is why the market-model specification is reported alongside the three-factor estimates. Under the market model, which contains no size factor, each friction event predicts −0.36 percentage points, significant at the 1% level, while watch-list status predicts +3.2 percentage points, significant at the 5% level.

[100]Akerlof (n 3)

[101]The friction count is heavily right-skewed, with most firms recording little or no friction; see Figure 8.

[102]A joint horse-race specification confirms that the two proxies load simultaneously without either absorbing the other, that adding sector and board-structure controls preserves both, and that governance variables carry no independent announcement-day effect; see Table 18 and Table 25.

Table 10: Friction and Announcement Returns (E1)

	FF3					CAPM
	(1)	(2)	(3)	(4)	(5)	(6)
Friction count	-0.0034***	-0.0031**	-0.0034**	-0.0031**	-0.0032**	-0.0036***
	(0.0013)	(0.0012)	(0.0016)	(0.0012)	(0.0016)	(0.0011)
Watchlist		0.0524*	0.0575	0.0512	0.0559	0.0323**
		(0.0316)	(0.0360)	(0.0327)	(0.0370)	(0.0127)
Above median		0.0227***	0.0211***	0.0175***	0.0203**	0.0263***
		(0.0062)	(0.0071)	(0.0067)	(0.0096)	(0.0058)
Illiquidity				-0.0018	-0.0014	
				(0.0098)	(0.0110)	
Catalist				-0.0098	-0.0001	
				(0.0126)	(0.0176)	
Financial controls	No	No	Yes	No	No	No
Sector FE	No	No	No	No	Yes	No
N	274	274	240	272	243	274
R^2	0.049	0.099	0.135	0.099	0.205	0.140

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1 (February 21, 2025). Columns (1)–(5): Fama–French three-factor model (Asia Pacific ex Japan). Column (6): market model (CAPM) with SGX equal-weighted index, presented jointly because the S$40 million watch-list threshold is a market-cap criterion and the FF3 SMB factor mechanically absorbs watch-list variation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Two decompositions identify what investors were pricing within the aggregate friction count. The first splits friction by type, distinguishing routine queries from the trading halts and suspensions that interrupt trading and from rarer formal regulatory actions (Table 11). Halts and suspensions drive the penalty entirely, with a coefficient of -0.0036 in the univariate specification, -0.0038 with channel controls, and -0.0036 with financial controls, significant at the 1% level in the first two specifications and at the 5% level in the third. Query-type friction is statistically indistinguishable from zero across all specifications, ranging from -0.0025 to -0.0013, while regulatory-action-type friction enters with a large negative coefficient between -0.013 and -0.043 but is too rare for reliable inference, since 97% of firms record no regulatory action. The market therefore priced the removal of oversight whose interruption of trading signals serious compliance problems, and not the removal of routine questioning, which indicates that what investors valued as informative monitoring was the severity of the regulatory signal rather than its frequency.

Table 11: Friction Type Decomposition (E1)

	(1)	(2)	(3)
Query-type friction	-0.0025	-0.0011	-0.0013
	(0.0023)	(0.0022)	(0.0025)
Halt/suspension-type friction	-0.0036***	-0.0038***	-0.0036**
	(0.0013)	(0.0012)	(0.0018)
Regulatory-type friction	-0.0127	-0.0236	-0.0433
	(0.0258)	(0.0275)	(0.0381)
Above median		0.0241***	0.0227***
		(0.0064)	(0.0073)
Watchlist		0.0627*	0.0818*
		(0.0360)	(0.0460)
Financial controls	No	No	Yes
N	274	274	240
R^2	0.056	0.112	0.148

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Query-type: SGX public queries. Halt/suspension-type: trading halts and suspensions. Regulatory-type: formal regulatory actions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The second decomposition splits friction by recency, separating events within the twelve months before E1 from older events, in order to ask whether friction operates as a flow signal about current compliance risk or as a stock of accumulated reputation (Table 12). Older friction drives the effect, with a coefficient that is marginally significant in the univariate specification at -0.0046 and remains marginally significant with channel controls at -0.0044, whereas recent friction is statistically indistinguishable from zero, at -0.0032 and -0.0026, and is if anything slightly less negative than the old-friction coefficient. The market thus treated regulatory attention as a stock of accumulated reputation rather than a flow of current scrutiny. The adverse-selection logic follows directly, since the firms penalised when the screening device is proposed for removal are those whose regulatory history the market had already incorporated into its pricing. Together, the two decompositions support reading the friction penalty as the loss of an informative monitoring signal rather than the removal of a current compliance burden.

Table 12: Friction Recency (E1)

	(1)	(2)
Friction count (recent)	-0.0032	-0.0026
	(0.0032)	(0.0030)
Friction count (old)	-0.0046*	-0.0044*
	(0.0027)	(0.0025)
Above median		0.0221***
		(0.0062)
Watchlist		0.0541*
		(0.0325)
N	274	274
R^2	0.060	0.108

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Recent friction: events from March 2024 to January 2025. Old friction: events before March 2024. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The cross-event behaviour of friction reinforces the reading. Pooling E1 and E3 in a stacked specification with event fixed effects, the E1 friction coefficient of approximately -0.003 is offset by an E3 interaction of $+0.003$, so the friction penalty at E3 is statistically indistinguishable from zero rather than reversing sign, in contrast to the size coefficient, which flips (Table 13). The size reversal is attributable to the mandate disclosure at E3, which changed expectations about where capital would be allocated, whereas the friction null reflects the absence of new friction-relevant information at that event, since E3 concerned capital deployment rather than oversight. The two channels accordingly exhibit distinct cross-event dynamics consistent with their reflecting different pricing dimensions: the size channel responds to the event that altered the mandate terms, while the friction channel is quiescent there. A generated-regressor bootstrap that accounts for the dependent variable being itself estimated leaves the friction coefficient marginally significant ($p = 0.054$), reflecting first-stage estimation noise that conventional inference does not capture, so the friction result is best read as robust in sign and economically stable, while its precision remains sensitive to the estimation of the underlying abnormal returns (Table 23). The penalty also rests on a comparatively rare form of regulatory action, since most firms carry little accumulated friction. That rarity is a further reason to treat the magnitude as indicative rather than sharply estimated, even as its sign and direction stay stable across the specification, type, and recency cuts.

Table 13: Stacked E1–E3 Friction Interaction

	(1)	(2)	(3)
Friction count	-0.0034***	-0.0029**	-0.0033**
	(0.0013)	(0.0012)	(0.0016)
Above median		0.0214***	0.0232***
		(0.0061)	(0.0071)
Post-E3	0.0345***	0.0572***	0.0557***
	(0.0065)	(0.0100)	(0.0112)
Friction × Post-E3	0.0040	0.0031	0.0044
	(0.0026)	(0.0025)	(0.0032)
Above median × Post-E3		-0.0387***	-0.0397***
		(0.0107)	(0.0110)
Financial controls	No	No	Yes
N	544	544	478
R^2	0.118	0.137	0.155

Notes: OLS with event fixed effects and standard errors clustered by stock code, because each firm contributes two observations (one at E1, one at E3); HC1 would understate the standard errors by treating these as independent, when in reality the same firm's CARs at E1 and E3 are correlated. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Post-E3 = 1 for E3 observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Targeted versus Broad Burden Removal

The information channel admits one systematic exception, which locates the design margin on this channel. Conditional on a firm's overall friction, watch-list status carried a positive coefficient at E1, marginally significant under the three-factor model at +5.2 percentage points and significant at the 5% level under the market model at +3.2 percentage points, indicating that the market valued removal of the watch-list label even as it penalised removal of general monitoring. The same deregulatory package therefore priced two legal objects in opposite directions: it treated the removal of accumulated regulatory history as a loss of monitoring and the removal of the watch-list label as relief from a financing burden. This is the pattern the information theory predicts once the watch-list is understood as an instrument whose stigmatising cost exceeds its informational content, so that its abolition retires a burden without discarding a signal.

A legal development that postdates the watch-list's introduction shapes the economic magnitude of the relief. Singapore enacted a court-supervised, Chapter 11-style restructuring regime in the Insolvency, Restructuring and Dissolution Act 2018, in force from 30 July 2020,[103] and the

[103] Insolvency, Restructuring and Dissolution Act 2018 (No 40 of 2018), in force 30 July 2020

availability of a functional restructuring path after 2020 makes removal of the watch-list trap more valuable than it would have been under the earlier regime: a distressed firm can now exit the self-reinforcing cycle of watch-list inclusion, financing impairment, and depressed valuation through formal restructuring rather than delisting.[104] This institutional complementarity between the abolition of the label and the prior creation of a restructuring alternative plausibly contributes both to the positive watch-list coefficient at E1 and to the larger gain at implementation documented below, since the value of escaping the trap depends on the existence of a viable exit other than delisting.

Although the comparison rests on only five treated firms and is best read as directional, the October implementation event sharpens the contrast between the two objects. When SGX RegCo formally abolished the watch-list on October 29, 2025, automatically removing all issuers, the five previously watch-listed firms in the sample gained 12.7 percent on average under the three-factor model, against 1.0 percent for other firms, with a similar pattern under the market model (7.3 percent versus −1.1 percent; Table 14, Panel A). The difference is economically large but imprecisely estimated given the small treated cell, and the Welch test does not reject equality of means ($p = 0.279$ under the three-factor model, $p = 0.363$ under the market model). The regression in Panel B confirms that watch-list status is the only cross-sectional variable predicting the E4 response, consistent with the event being narrowly targeted at watch-list removal rather than conveying broader information; the watch-list coefficient does not, however, survive inference that accounts for the small cell ($p = 0.30$ under the three-factor model). The Panel B magnitudes are therefore reported as illustrative rather than precisely estimated.

[104]Meng Seng Wee and Hans Tjio, 'Singapore as International Debt Restructuring Centre: Aspiration and Challenges' (2022) 57 Tex Intl LJ 1

Table 14: Watch-List Removal Implementation (E4)

Panel A: Mean CARs by watchlist status

	FF3		CAPM	
	Mean CAR	N	Mean CAR	N
Watchlisted	0.1273	5	0.0731	5
	(0.0939)		(0.0819)	
Non-watchlisted	0.0098***	253	-0.0111***	253
	(0.0036)		(0.0035)	
Difference	0.1176		0.0841	
	(1.25)		(1.03)	

Panel B: Regression

	FF3		CAPM	
	(1)	(2)	(3)	(4)
Watchlist	0.1176	0.1110	0.0841	0.0817
	(0.0844)	(0.0847)	(0.0737)	(0.0740)
Above median		-0.0129*		-0.0048
		(0.0072)		(0.0071)
N	258	258	258	258
R^2	0.063	0.073	0.036	0.038

Notes: CARs estimated over the $(-1, +3)$ trading-day window around E4 (October 29, 2025). Panel A: Mean CARs under both FF3 (Fama–French three-factor, Asia Pacific ex Japan) and CAPM (market model, SGX equal-weighted index). Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses for group means; significance is assessed by the Student t-test. Welch t-statistic in parentheses for the difference. Panel B: OLS with HC1 robust standard errors in parentheses. Columns (1)–(2): FF3. Columns (3)–(4): CAPM. CAPM presented jointly because the S$40 million watch-list threshold is a market-cap criterion and the FF3 SMB factor mechanically absorbs watch-list variation. Panel A includes all firms with valid CARs ($N = 263$); Panel B regression sample is $N = 258$ after excluding firms with missing control variables. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The two pieces of information-channel evidence carry different weight, and the distinction governs how the channel's claims are supported. The friction results, estimated on the full cross-section, are the workhorse, because every claim about the information channel rests on them: that broad oversight removal was priced as information loss, that the penalty concentrates in severe and historical scrutiny rather than routine or recent scrutiny, and that it is independent of size. The watch-list results illustrate a further and narrower point – that burden removal can be made selective, retiring a stigmatising label while leaving informative monitoring in place – but they rest on five treated firms. The paper accordingly treats them as corroborative and draws no information-channel conclusion that would fail if the watch-list evidence were set aside entirely. Taken together, the cross-sectional patterns track the specific content of each instrument rather than the fact of intervention, which is the premise the design implications develop.

6 Regulatory Design Implications

The findings recast a familiar policy disappointment as a problem of design. Programmes that deploy public capital into equity markets and that liberalise listing oversight are routinely judged by their scale, yet the evidence here indicates that scale did not determine whether Singapore's reform reached its intended firms. The market priced the same S$5 billion commitment first as a large-cap event and then, five months later, as a small-cap event once named managers and specified allocation terms accompanied it, and it priced the same deregulatory programme as a loss for monitored firms and as a relief for stigmatised ones according to which instrument was withdrawn. A general lesson about regulatory architecture follows: interventions delegated through institutional channels succeed or fail on the terms of the delegation rather than on the size of the commitment, which is one instance of the broader mismatch between regulatory instruments and the systems they are meant to govern – the gap that this article labels the 'regulatory intermediation gap'.[105] In what follows, we develop the design margin on each channel and then locate the conditions under which the gap is most likely to bind.

6.1 Designing the Capital Mandate

On the capital channel, the operative design variable is the specificity of the mandate that governs how delegated capital is deployed – hereafter, 'mandate specificity'. A commitment of public money to professional managers does not by itself determine where the money goes, because the managers' investability constraints favour large and liquid securities, so the commitment reaches an underserved segment only when the mandate constrains that default tilt. The reform makes the point visible in its own timeline. The February commitment named no managers and disclosed no allocation terms, and the market priced it as a large-cap event; the July appointment of named managers carrying mandate terms that specified significant allocation to small- and mid-cap equities reversed the size gradient. Because the two events hold the size of the appropriation fixed and vary only the specificity of the terms attached to it, the reversal isolates the terms, not the money, as the variable the market repriced.

Singapore's reform provides a concrete example of what a workable allocation expectation can comprise, though the components it combined are better read as features that plausibly reinforced credibility than as individually necessary conditions. The first component is a stated segment target that an outside observer can evaluate against a portfolio: the public formulation that appointed managers make a "significant allocation" to small- and mid-cap equities ([EQDP-TERMS]) directs capital toward a defined class of firms rather than toward SGX-listed equities at large. The second is the identification of the managers who carry the expectation, since naming Avanda, Fullerton, and J.P. Morgan Asset Management attaches the target to specific fiduciaries whose conduct can be observed and compared, which converts a programme-level aspiration into an obligation borne by identifiable agents. The credibility of that attachment is reinforced by the managers' provenance, since Avanda Investment Management was founded by former GIC officers and Fullerton Fund

[105] Dan Awrey and Kathryn Judge, 'Why Financial Regulation Keeps Falling Short' (2020) 61 BC L Rev 2295

Management is Temasek-linked, both read by the market as state-aligned investors committed to Singapore's domestic market.[106] The third is disclosure of the terms relative to the commitment, so that the market learns not only that capital has been committed but on what conditions it will be deployed; until the terms are public, investors price the unconstrained manager rather than the constrained one. A fourth component, which the reform left implicit rather than specified, is accountability and monitoring. An allocation expectation that no one observes ex post and that carries no consequence for deviation is priced as cheap talk, whereas the prospect that deployment will be measured against the stated target makes the expectation binding in the only sense the market cares about – its effect on where the marginal dollar is expected to go. The timing margin then governs how quickly the constraint is impounded. Because the credible information arrived at the manager-appointment stage rather than at the original commitment, the months between February and July were months in which the market priced the programme as it would price any unconstrained mandate, and the reversal coincided with the disclosure of terms rather than with the arrival of any new appropriation. Sequencing is therefore itself a design choice, in that a programme committing capital well before it specifies the terms invites a sustained period of mispricing against its own objective.

The comparison with other jurisdictions' programmes clarifies why this configuration is distinctive (Table 15). The instruments differ in the channel they engage and in whether they attach a segment-specific allocation expectation to the deploying intermediary. The Bank of Japan's exchange-traded-fund programme deployed central-bank capital on a far larger scale, but it operated through passive replication of broad indices executed by an appointed trust bank, so its incidence fell on firms already large enough to secure index inclusion rather than on an underserved segment.[107] The contrast is instructive precisely because the BOJ channel is the closest precedent for routing public capital into listed equities: it shows that the channel form – passive index replication for a monetary-policy objective – fixes the distribution of benefits independently of scale, since the intervention can reach a firm only by reaching the index that contains it. Korea's Corporate Value-up Programme engages the issuer side rather than the manager side, working through voluntary disclosure of value-up plans, governance incentives, and an index whose construction screens on market capitalisation and so tilts toward larger constituents, without a capital mandate directing managers toward smaller firms;[108] its headline gains accordingly concentrated among larger issuers, which is what a design that supplies recognition and infrastructure but no segment-specific allocation expectation would predict. China's Government Guidance Funds route public capital through appointed managers, as the EQDP does, but predominantly into venture-capital and private-equity vehicles rather than secondary-market purchases of listed equities,[109] so the intermediary constraints they confront are those of private-market deployment rather than of benchmark-relative public-equity management. The European Union's Listing Act lowers the regulatory cost of being listed without deploying capital at all,[110] addressing the supply of listings rather than the flow of capital toward existing small-caps. Read across the matrix, the programmes vary the channel and the target but hold mandate specificity roughly constant, and none pairs

[106] [CITE-NEEDED: provenance of Avanda (founded by former GIC officers) and Fullerton (Temasek-linked)].
[107] Charoenwong and others (n 30); Katagiri and others (n 30)
[108] Lee and Peh (n 29)
[109] Huang and others (n 26)
[110] the Listing Act (n 27)

an active, direct-purchase mandate with a disclosed segment-specific allocation expectation. The EQDP occupies that cell alone, and the cross-sectional reversal indicates that this combination – and specifically the disclosure of the terms – is what allowed intermediated capital to be priced as reaching the intended segment.

The implication for programme design is that the mandate functions as the operative policy instrument, while the size of the appropriation mainly signals ambition. The market prices the terms that govern the intermediary, so a programme that announces capital without specifying allocation invites the intermediation gap that the February announcement displayed. This does not make specificity costless, and the design problem is genuinely two-sided. Cumulated across the two events the gap narrowed without closing, which is what one would expect if mandate terms move capital toward the target segment while leaving the managers some discretion over which firms within it to hold. That residual discretion is valuable. Fund-manager caution toward thin, illiquid, and informationally opaque firms is partly a rational response to real risk differentials, and a mandate stringent enough to override it – by fixing allocation shares tightly or by removing the manager's latitude to decline the weakest names – would reintroduce precisely the risks that delegation to professional managers is meant to manage. The design objective is accordingly to specify terms that redirect capital toward the target segment while preserving the intermediary's discretion to price genuine risk, rather than to maximise either the size of the commitment or the stringency of the allocation expectation taken on its own. A credible expectation that constrains the portfolio tilt without dictating the security selection occupies the interior of that trade-off, and the partial narrowing of the gap is consistent with the reform having reached it.

6.2 Information-Preserving Deregulation

On the information channel, the operative design variable is the selectivity of burden removal. The deregulatory programme bundled two kinds of instrument that the market priced in opposite directions: mechanisms that produce information about firm quality, such as public queries, trading halts, and suspensions, and mechanisms that impose a burden in excess of the information they convey, such as the stigmatising watch-list label. Removing the first was priced as information loss for the firms whose quality the market had been inferring from regulatory attention, while removing the second was priced as relief for the firms the label had impaired. A deregulatory package that does not distinguish the two retires useful signals together with genuine burdens, and the differential pricing across high-friction firms indicates that the distinction is one investors can draw and do draw. The selectivity of removal, captured here by the contrast between the friction results and the priced relief from watch-list abolition, is what separates deregulation that degrades price formation from deregulation that lifts a deadweight cost.

The point bears directly on the choice SGX RegCo made in shifting from public queries to private engagement, which is the cleanest instance in the reform of a load-bearing public signal being withdrawn. A public query is an informational instrument as well as a supervisory one, because the fact of the query, once disclosed, tells the market that the regulator has identified a concern at a particular firm, so the signal is produced by the publicity of the instrument rather than by the supervision behind it. Replacing the public query with private engagement can leave the underlying supervision entirely intact while removing the signal, since the regulator continues to investigate but the market no longer observes that it has. The shift is therefore a change in what is disclosed rather than in what is supervised, and its informational cost is borne by the price-formation process even if the conduct of oversight is unchanged or improved. A regulator that retires a public oversight instrument in this way faces a design question: what, if anything, substitutes for the information the instrument had been producing.

A substitute public signal is conceivable, and naming its candidates clarifies the design space. The regulator could publish a periodic record of the firms with which it has engaged privately, disclose engagement on a lag once a matter is resolved, or require the firm itself to announce that it has received a regulatory enquiry, each of which would restore some of the informational content of the public query while softening the immediacy that made the query stigmatising. The most general substitute is issuer disclosure itself, since a disclosure-based regime presumes that what issuers are obliged to reveal can carry the informational load that public oversight had carried. That presumption holds only under demanding conditions. Disclosure reaches and informs investors when it is salient, comprehensible, and acted upon, and the experimental evidence on disclosure efficacy shows that these conditions frequently fail, particularly for unsophisticated recipients who do not fully incorporate disclosed information into their decisions.[111] The disclosure-efficacy condition is consequential for markets with a substantial retail base, such as Singapore, for which a regulator-supplied public signal such as the query may do informational work that voluntary issuer disclosure does not replicate; the concern is sharper where the investor-compensation and recourse regime remains thin, as Singapore's does even as it is under review.[112] The distinction that the design must respect is accordingly between removing a stigmatising label, whose costs to the firm exceeded its informational content and whose retirement the market welcomed, and removing informative monitoring, whose signal the market had been using and for which it priced no automatic substitute. Information-preserving deregulation therefore requires the regulator to identify which of its instruments are load-bearing for price formation and to supply a substitute signal where a public instrument is withdrawn, rather than treating all oversight as undifferentiated burden.

6.3 Boundary Conditions and the Integrated Regulator

The gap documented here is not a universal feature of intermediated capital programmes, and its design lessons bind most tightly under three identifiable institutional conditions. The first is a skewed issuer size distribution that produces a discrete investability threshold rather than a smooth gradient, so that the intended beneficiaries sit below the level at which professional managers will transact at all and a fund manager faces a near-binary inclusion decision rather than a continuous schedule of portfolio costs. The second is a concentrated intermediary structure,

[111] Geneviève Helleringer, 'Designing Disclosures: Testing the Efficacy of Disclosure in Retail Investment Advice' in Klaus Mathis and Avishalom Tor (eds), *Nudging — Possibilities, Limitations and Applications in European Law and Economics* (Springer 2016) 153

[112] [CITE-NEEDED: Singapore investor-compensation/recourse regime, still thin and under review].

in which a small number of managers control the marginal public-capital allocation decision, so that their shared constraints move prices in the same direction rather than washing out across heterogeneous strategies. The third is a reliance on publicly observable oversight signals, because the information channel operates only where investors can see the instruments whose removal they then price; where supervision is conducted privately, there is no public signal to withdraw and the information-loss effect cannot arise. Singapore satisfies all three conditions, which makes it a clean setting for the test and at the same time marks the boundary of the result.

Stating the conditions as variables shows where the gap should concern other regulators and where it should not. It should weaken in deep, liquid markets with continuous size distributions and broad analyst coverage, because the investability threshold blurs into a gradient and professional managers face a smooth schedule of portfolio costs rather than a discrete inclusion decision, so that a capital programme works with the existing distribution of attention rather than against a cliff in it. It should likewise weaken where intermediation is dispersed across many competing managers with heterogeneous mandates, because no single constraint then dominates the marginal allocation and competition attenuates the common tilt. By the same logic the gap should bind in settings that share Singapore's conditions: smaller European exchanges and growth boards with thin coverage and a pronounced size cliff, Hong Kong's GEM and comparable second boards where small issuers cluster below the level at which institutional capital transacts, and mid-tier Asian markets with concentrated intermediation and publicly observable oversight regimes. The information channel travels on its own condition. Where regulatory oversight runs through public announcements that investors price, as in Singapore, the withdrawal of those announcements is priced as information loss; where supervision runs through private channels with few public signals, the same deregulation should be muted because there is little observable signal to remove. The gap is therefore most likely to arise where governments route corrective capital through a concentrated set of professional intermediaries into markets characterised by sharp investability cliffs and visible oversight regimes, and it should attenuate as any of those conditions relaxes. Whether the gap responds to mandate specificity in deeper markets, where these conditions hold only in attenuated form, is the natural question for further work.

These conditions also bear on the architecture that produced the reform. Singapore's integrated regulatory architecture issued a developmental instrument through MAS and a supervisory instrument through SGX RegCo within a single package, and the evidence indicates that the two operate through different transmission economics, the capital channel through intermediary investability constraints and the information channel through the structure of public signals. An architecture that houses both functions therefore confronts two distinct design problems rather than one, and the solutions the evidence supports differ across them, with mandate specificity addressing the developmental side and selectivity of removal preserving information on the supervisory side. This concerns the design tasks such an architecture sets, not the bodies' internal deliberations. How comparable design problems are handled where the developmental and supervisory functions sit in separate bodies, as under a twin-peaks architecture, is a question that the standard treatments of regulatory architecture frame but that this paper does not resolve,[113] and we leave it to future work in jurisdictions running comparable programmes.

[113] John Armour and others, *Principles of Financial Regulation* (Oxford University Press 2016)

7 Conclusion

This article has examined Singapore's 2025 reform, an intervention that committed public capital and liberalised oversight within a single regulatory architecture, with MAS committing the capital and SGX RegCo recalibrating oversight. The capital instrument was directed at small- and mid-cap firms, while the supervisory reforms reached the firms carrying the heaviest regulatory history; the market priced both as likely to miss the firms they were meant to help, and in each case the shortfall traced to the institutional channel through which the policy was transmitted rather than to the scale of the commitment. Capital routed through professional managers was priced as a large-cap event until the disclosed mandate terms specified allocation to the target segment, and broad oversight removal was priced as information loss, the market separating the withdrawal of a stigmatising label, which it treated as relief, from the withdrawal of informative monitoring, which it treated as loss. The divergence between regulatory intent and market outcome that we have termed the regulatory intermediation gap is therefore a property of design, and it responds to design: the July mandate disclosure narrowed the gap on the capital channel, while on the information channel the selectivity of removal, illustrated by the targeted watch-list abolition, determined whether deregulation was priced as relief or as loss.

The lesson generalises beyond Singapore to the broad class of interventions that deploy public capital through private intermediaries and that recalibrate public oversight, because in both the policy must pass through a channel whose own constraints shape the result. Where governments route corrective capital through the same intermediaries whose constraints produced the capital deficit, and where they withdraw the public signals investors had used to price opaque firms, the gap is generally a predictable consequence unless the terms of delegation and the selectivity of removal are designed against it. The conditions under which the gap binds most tightly – a skewed size distribution, concentrated intermediation, and publicly observable oversight – mark where these design lessons travel and where they weaken.

For the regulator, the practical implication is that the instruments of delegation deserve the attention usually reserved for the size of the programme. A mandate specific enough to bind an intermediary, and a burden removal selective enough to preserve information, are modest design choices that carry first-order effects on whether an intervention reaches the firms it targets. Whether the gap responds to design in the same way in deeper and more liquid markets, where the institutional conditions hold in attenuated form, remains an open empirical question that the continued spread of comparable programmes should allow future work to address. The design margin this paper isolates, namely the specificity of the terms attached to delegated authority, recurs wherever a government pursues a developmental objective through private intermediaries or recalibrates a public oversight regime, so the questions the reform makes tractable reach well beyond the equity-market setting in which they are posed here.

Table 1: Actor–Authority Map of Singapore's 2025 Equity-Market Reform

Event	Date	Acting body	Instrument	Statutory / regulatory basis
E0	2 Aug 2024	MAS	Formation of the Equities Market Review Group (no substantive measures)	MAS developmental mandate ([MAS-STAT-CITE])
E1	21 Feb 2025	MAS	S$5bn EQDP capital commitment; investor-participation measures (GIP Option C, GEMS, tax exemptions)	MAS developmental mandate; EQDP capital commitment ([MAS-STAT-CITE])
(E1)	(cont.)	SGX RegCo	Proposed disclosure-based regime: listing-review consolidation, streamlined admission and prospectus requirements, watch-list removal proposal	Delegated listing-regulation authority; proposed delegation of prospectus lodgment / registration ([MAS-STAT-CITE])
E2	15 May 2025	SGX RegCo	Consultation paper on a more disclosure-based regime (watch-list removal, query-to-engagement shift, narrowed suspension criteria)	Delegated listing-regulation authority
E3	21 Jul 2025	MAS	Appointment of first three EQDP managers; S$1.1bn deployed; disclosure of small/mid-cap mandate terms; investor-recourse proposals	MAS developmental mandate; EQDP mandate terms ([EQDP-TERMS])
E4	29 Oct 2025	SGX RegCo	Watch-list abolition; cessation of public trading queries (replaced by private engagement); listing-review rule changes	Delegated listing-regulation authority; listing-rule amendment power
E5	19 Nov 2025	MAS	Final Review Group report; second batch of six EQDP managers (S$2.85bn); SGX–Nasdaq dual-listing bridge	MAS developmental mandate; EQDP capital commitment ([MAS-STAT-CITE])
(E5)	(cont.)	SGX RegCo	Market-structure measures (board-lot reduction, designated-market-maker incentives)	Exchange market-operation and listing-rule authority

Notes: The map distinguishes the developmental instruments, committed and deployed by the Monetary Authority of Singapore (MAS) under its developmental mandate (the EQDP capital at E1 and the manager appointments and mandate disclosure at E3 and E5), from the supervisory instruments, issued by Singapore Exchange Regulation (SGX RegCo) under its delegated listing-regulation authority and MAS's statutory oversight (the consultation, deregulation, and watch-list abolition at E1, E2, E4, and E5). The Equities Market Review Group (E0, E1, E5) was chaired by the Deputy Chairman of MAS. E0 is excluded from hypothesis testing. Placeholder tokens [MAS-STAT-CITE] and [EQDP-TERMS] mark the precise statutory provisions and mandate terms to be supplied.

Table 15: Comparative Design of Intermediated Equity-Market Programmes

Programme	Channel	Mandate specificity	Target segment	Disclosure timing
Singapore EQDP (2025)	Active mandates to professional fund managers for direct secondary-market purchase of listed equities	Segment-specific: appointed managers expected to make "significant allocation" to small- and mid-cap stocks ([EQDP-TERMS])	Small- and mid-cap SGX-listed equities	Capital committed at announcement (E1, Feb 2025); mandate terms and named managers disclosed later (E3, Jul 2025)
Korea Corporate Value-up (2024)	Voluntary issuer disclosure plus governance incentives and index-linked infrastructure; no direct capital mandate to managers	Issuer-directed, not segment-targeted: firms voluntarily publish value-up plans; no allocation mandate	Listed issuers broadly; the Value-up Index tilts toward larger constituents	Issuer plans disclosed on a voluntary, recurring basis; index launched Sep 2024
Bank of Japan ETF programme (2010–2024)	Passive index replication; purchases executed by an appointed trust bank holding ETFs as trust property	Not a segment-targeting mandate: purchases track broad equity indices for monetary-policy purposes	Index-included (typically larger) firms via TOPIX / Nikkei 225 ETFs	Purchase terms and indices published as monetary-policy operating terms; benefits accrue through index membership
China Government Guidance Funds	Public capital routed through appointed investment managers, primarily venture-capital and private-equity vehicles	Mandate scope varies by fund; oriented to strategic and early-stage sectors rather than listed small-caps	Primary and private-market firms, not secondary-market listed equities	Fund mandates set at establishment; not an announcement-based listed-equity intervention
EU Listing Act (2024)	Supply-side market-structure reform; no public capital deployed through intermediaries	Not an allocation mandate: eases listing and disclosure requirements to lower barriers to public markets	Small and medium-sized enterprises seeking listing	Entered into force Dec 2024; operates through statutory rule change, not periodic mandate disclosure

Notes: The table contrasts intermediated equity-market programmes along the design margins that govern whether public support reaches an intended segment. "Channel" records how the programme transmits support; "Mandate specificity" records whether the instrument attaches a segment-specific allocation expectation to the deploying intermediary; "Target segment" records the firms the programme is configured to reach; and "Disclosure timing" records when the operative terms become public relative to commitment. The Singapore EQDP is distinctive in pairing an active, direct-purchase mandate with a segment-specific allocation expectation, with the operative terms disclosed at E3 after capital was committed at E1. The Bank of Japan programme transmits support through passive index replication, so its incidence is determined by index membership rather than by a targeting mandate. Korea's

A Robustness and Supplementary Results

This appendix gathers the identification tests, the robustness checks, and the supplementary event-level results that the main text relies on but does not display.

A.1 Identification and Placebo Tests

A.1.1 Paired Reversal Placebo

The E1-to-E3 reversal in the size coefficient supplies much of the evidence for the intermediation gap hypothesis, so the threshold question is whether a reversal of that magnitude could arise by chance. To address it, we construct a paired-date reversal placebo. Two hundred pairs of pseudo-event dates are sampled uniformly from June 2023 to January 2025, each pair separated by approximately 105 trading days – the span that matches the E1–E3 gap – and excluding the window around E0. For each pair, firm-level CARs are estimated at both dates using a standard $[-250, -30]$ estimation window, and the size and friction coefficients from the channel-separation horse race in Table 18 are computed. The reversal statistic $\Delta\hat{\beta} = \hat{\beta}_{\text{date}_2} - \hat{\beta}_{\text{date}_1}$ records the change in each coefficient across the pair.

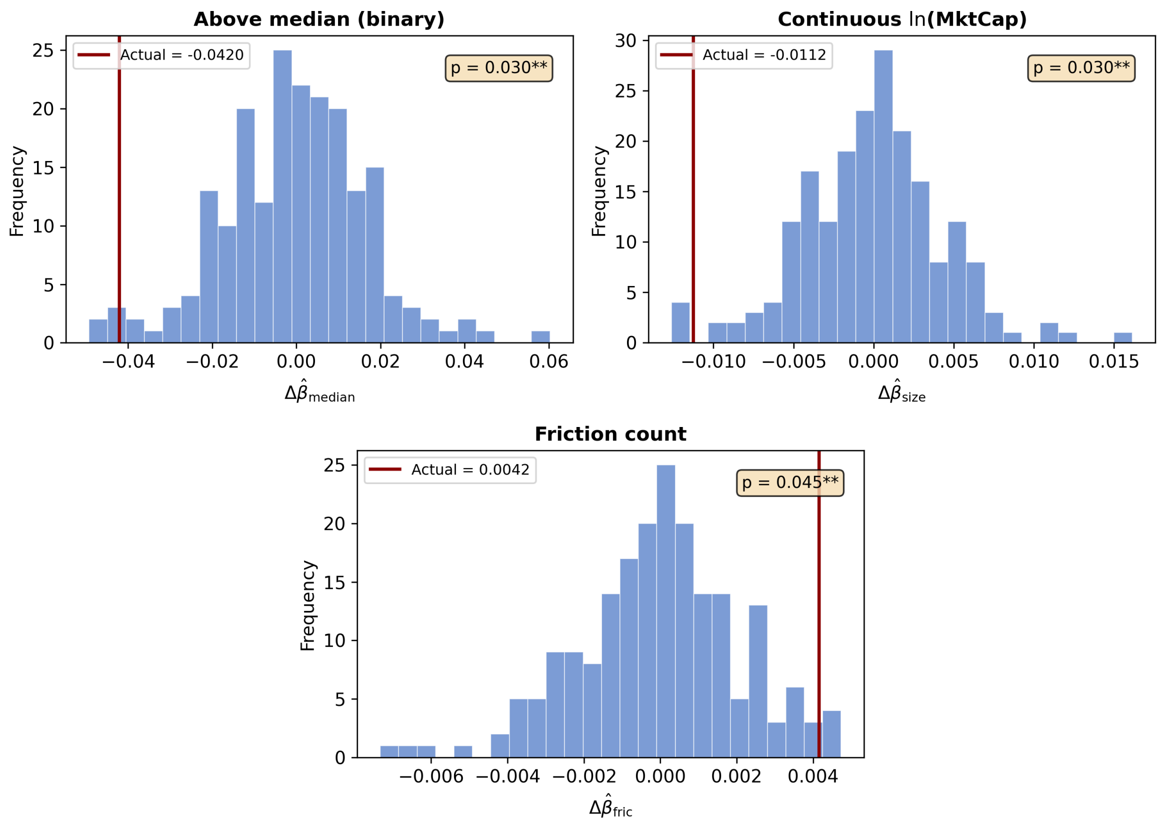
Table 16 places the binary above-median reversal at the 2nd percentile of the placebo distribution, with $\Delta\hat{\beta}_{\text{median}} = -0.042$ ($p = 0.030$); only 6 of the 200 placebo pairs generate a reversal of comparable magnitude. The continuous size measure reversal ($\Delta\hat{\beta}_{\text{size}} = -0.011$, 2nd percentile, $p = 0.030$) is similarly far into the tail. Both size reversals reject the null that the E1-to-E3 sign flip could arise from random cross-sectional variation. The friction coefficient reversal ($\Delta\hat{\beta}_{\text{fric}} = 0.004$, 98th percentile, $p = 0.045$) is also significant at the 5% level, which indicates that the information channel, like the capital channel, follows an atypical cross-event pattern. Figure 6 plots the three placebo distributions with the observed reversals marked, and in each panel the actual value sits in the extreme tail.

Table 16: Paired Reversal Placebo (200 Pseudo-Event Pairs)

Variable	Actual $\Delta\hat{\beta}$	Placebo Mean	Placebo SD	Pctile	Perm. p
Above median	−0.0420	−0.0007	0.0169	2.0%	0.030**
log(Market cap)	−0.0112	−0.0001	0.0044	2.0%	0.030**
Friction count	+0.0042	−0.0000	0.0021	98.0%	0.045**

Notes: 200 pseudo-event date-pairs separated by ≈105 trading days (matching the E1–E3 gap). $\Delta\hat{\beta} = \hat{\beta}_{\text{date}_2} - \hat{\beta}_{\text{date}_1}$ from the channel-separation horse race (Table 18) with HC1 robust standard errors. Two-sided p: fraction of placebos with $|\Delta\hat{\beta}| \geq |\Delta\hat{\beta}_{\text{actual}}|$. Above median = 1 if $\ln(\text{MktCap})_i \geq$ sample median. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 6: Distribution of placebo coefficient reversals from 200 random date-pairs. Vertical lines mark the observed E1-to-E3 reversals.



Note: Top-left panel: distribution of $\Delta\hat{\beta}_{\mathrm{median}}$ (binary above-median coefficient) across 200 placebo pairs; the actual reversal (-0.0420, maroon line) falls at the 2nd percentile ($p = 0.030$). Top-right panel: distribution of $\Delta\hat{\beta}_{\mathrm{size}}$ (continuous ln(Market cap) coefficient); the actual reversal (-0.0112) falls at the 2nd percentile ($p = 0.030$). Bottom panel: distribution of $\Delta\hat{\beta}_{\mathrm{fric}}$ (friction count); the actual reversal (0.0042) falls at the 98th percentile ($p = 0.045$). All three reversals are statistically significant at the 5% level.

A.1.2 Cross-Market Placebo: Hong Kong

The logic of the Hong Kong placebo is to ask whether the size gradient appears in a neighbouring market that did not experience the reform. The cross-market placebo underlying Figure 4 matches each SGX firm to its three nearest HKEX neighbours by log market capitalisation ($K = 3$), which yields a matched sample of 644 Hong Kong firms, and the identical size–CAR specification is then estimated on both markets at E1. Panel A of Table 17 reports both the continuous $\ln(\mathrm{MktCap})_i$ specification and the binary above-median indicator. Under the continuous measure the SGX coefficient is positive and significant at the 1% level (0.006), whereas the matched HK coefficient is negative and marginally significant (-0.004), so the cross-market difference is itself significant ($p = 0.001$); the binary indicator sharpens the same contrast to the 1% level. Panel B turns to size quartile CARs within each market. The SGX gradient rises monotonically from $Q1$ to $Q4$, while the HK sample runs the other way, posting a significantly negative gradient whose $Q4$–$Q1$ spread of −2.78 percentage points ($p = 0.024$) is opposite in sign to the SGX effect.

Table 17: Cross-Market Placebo: SGX vs Matched Hong Kong Sample (E1)

	SGX	HK (matched)	Difference	p
Panel A: Size regressions (E1, FF3 CAR[−1, +3])				
log(Market cap)	+0.0057***	−0.0040*	+0.0097***	0.001
	(0.0020)	(0.0021)	(0.0029)	
Above median	+0.0248***	−0.0166*	+0.0414***	0.000
	(0.0071)	(0.0088)	(0.0113)	
N	272	644	916	
R^2	0.035	0.005		

Panel B: Size quartile CARs (%, E1)

	SGX		HK (matched)	
	Mean CAR	N	Mean CAR	N
Q1 (smallest)	−2.75***	68	−0.41	161
Q2	−2.33***	68	−1.96**	161
Q3	−0.34	68	−2.51***	161
Q4 (largest)	+0.22	68	−3.19***	161
$Q4 - Q1$	+2.98***		−2.78**	
	[$t = 2.74$]		[$t = -2.26$]	
	($p = 0.007$)		($p = 0.024$)	
Cross-market $Q4$−$Q1$ difference: +5.75 pp*** ($p = 0.000$)				

Note: Panel A reports OLS regressions of FF3 CARs $(-1, +3)$ on size measures. The HK sample consists of 644 HKEX firms matched to SGX firms by log market capitalisation ($K = 3$ nearest neighbours). HC1 robust standard errors in parentheses. The difference column tests H_0: $\beta_{\text{SGX}} = \beta_{\text{HK}}$ using $z = (\hat{\beta}_{\text{SGX}} - \hat{\beta}_{\text{HK}})/\sqrt{SE^2_{\text{SGX}} + SE^2_{\text{HK}}}$. Panel B reports mean CARs by size quartile within each market. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2 Specification Robustness

A.2.1 Channel-Separation Horse Race

At E1 the two channel proxies load jointly without either absorbing the other, which is the empirical basis for treating capital and information as separate pricing dimensions. Table 18 reports the horse race with controls added in steps: entered jointly, the above-median size indicator (0.018) and the friction count (−0.0031) each remain significant, and adding sector fixed effects and board-structure controls preserves both, while board independence and CEO-chair separation carry no independent announcement-day effect, consistent with the univariate governance regressions across events (Table 25). The size–friction correlation is only $\rho = -0.21$ (Figure 7), so the joint test is meaningful rather than mechanically determined.

Table 18: Channel Separation: Joint Specification (E1)

	(1)	(2)	(3)	(4)	(5)
Above median	0.0249***		0.0175***	0.0203**	0.0193***
	(0.0065)		(0.0067)	(0.0096)	(0.0072)
Friction count		-0.0034***	-0.0031**	-0.0032**	-0.0030**
		(0.0013)	(0.0012)	(0.0016)	(0.0013)
Watchlist			0.0512	0.0559	0.0508
			(0.0327)	(0.0370)	(0.0329)
Illiquidity (z-Amihud)			-0.0018	-0.0014	-0.0012
			(0.0098)	(0.0110)	(0.0099)
Catalist			-0.0098	-0.0001	-0.0055
			(0.0126)	(0.0176)	(0.0129)
Independence share					-0.0195
					(0.0244)
No CEO duality					0.0085
					(0.0102)
Sector FE	No	No	No	Yes	No
N	274	274	272	243	260
R^2	0.051	0.049	0.099	0.205	0.096

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1 (February 21, 2025), estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Above median = 1 if $\ln(\text{MktCap})_i \geq$ sample median. Spec (4) includes SSIC level-1 sector fixed effects. Spec (5) adds board independence share and CEO-chair separation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.2 CAR Window Robustness

Table 19 re-estimates the E1 horse race specification across five CAR windows. The above-median size indicator is significant at the 5% level in all five, ranging from 0.011 at $(-1, +1)$ to 0.025 at $(-1, +5)$. This uniformity, taken together with the wider confidence intervals around the continuous $\ln(\text{MktCap})_i$ measure reported in Section A.2.5, supports the threshold reading: the size effect behaves as a binary investability distinction rather than a smooth gradient. The friction coefficient is likewise stable across windows, at -0.0031 for the baseline $(-1, +3)$, -0.0034 at

$(-1,+5)$, and -0.0027 at $(0,+2)$, each significant at the 5% level. The narrowest window $(-1,+1)$ yields a smaller friction coefficient (-0.0017) that remains significant at the 5% level, which suggests that friction-related information was impounded over several trading days rather than on the announcement day alone.

Table 19: CAR Window Robustness: E1 Horse Race

	$(-1,+1)$	$(-1,+3)$	$(-1,+5)$	$(0,+2)$	$(0,+5)$
Above median	0.0112**	0.0175***	0.0250***	0.0112**	0.0214**
	(0.0052)	(0.0067)	(0.0089)	(0.0051)	(0.0083)
Friction count	-0.0017**	-0.0031**	-0.0034**	-0.0027**	-0.0024*
	(0.0008)	(0.0012)	(0.0016)	(0.0011)	(0.0015)
Watchlist	0.0282	0.0512	0.1111**	0.0610*	0.0914**
	(0.0192)	(0.0327)	(0.0485)	(0.0325)	(0.0444)
Illiquidity	0.0023	-0.0018	-0.0076	-0.0088	-0.0088
	(0.0038)	(0.0098)	(0.0112)	(0.0100)	(0.0118)
Catalist	-0.0182*	-0.0098	-0.0063	-0.0028	-0.0012
	(0.0100)	(0.0126)	(0.0141)	(0.0094)	(0.0124)
N	267	272	274	269	274
R^2	0.099	0.099	0.101	0.113	0.076

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the indicated trading-day window around E1, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Above median = 1 if $\ln(\text{MktCap})_i \geq$ sample median. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.3 Factor Model Robustness

Table 20 estimates the E1 horse race specification separately under CAPM, FF3, and the Fama–French five-factor model (FF5).[114] The above-median coefficient is significant at the 1% level under CAPM (0.020) and FF3 (0.018), and at the 5% level under FF5 (0.020). The friction coefficient stays negative and stable, at -0.0036 under CAPM, where it is significant at the 1% level, and -0.0031 under FF3 and -0.0029 under FF5, each significant at the 5% level. The CAPM friction coefficients run approximately 16% larger than their FF3 counterparts, which reflects the Asia Pacific ex Japan SMB factor absorbing part of the friction effect and the expected overlap between size and friction. The watch-list coefficient behaves less robustly: significant at the 5% level under CAPM (0.030), it attenuates to insignificance under FF3 (0.051) and FF5 (-0.012), again because SMB absorbs the relevant variation. The core size and friction results hold across all three factor models.

[114]Eugene F Fama and Kenneth R French, 'A Five-Factor Asset Pricing Model' (2015) 116 J Fin Econ 1

Table 20: Factor Model Robustness: E1 Horse Race

	CAPM	FF3	FF5
Above median	0.0204***	0.0175***	0.0198**
	(0.0064)	(0.0067)	(0.0086)
Friction count	-0.0036***	-0.0031**	-0.0029**
	(0.0011)	(0.0012)	(0.0013)
Watchlist	0.0297**	0.0512	-0.0122
	(0.0137)	(0.0327)	(0.0318)
Illiquidity	-0.0015	-0.0018	0.0281***
	(0.0030)	(0.0098)	(0.0059)
Catalist	-0.0124	-0.0098	-0.0081
	(0.0115)	(0.0126)	(0.0162)
N	272	272	234
R^2	0.143	0.099	0.116

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1. CAPM: market model with SGX equal-weighted index. FF3: Fama–French three-factor model (Asia Pacific ex Japan, primary specification). FF5: Fama–French five-factor model (Asia Pacific ex Japan). The CAPM friction coefficient is ≈16% larger than FF3 because the APXJ SMB factor absorbs part of the friction–size overlap. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.4 Size Effect Across Alternative Measures

Panel A of Table 21 reports univariate size regressions by event. The E1 coefficient is significant at the 1% level (0.0054) and the E3 coefficient is significant at the 5% level (-0.0059), a sign flip that is consistent with Bayesian learning as the market updated once mandate details were disclosed. Panel B then carries the reversal across four alternative size measures – log market cap, log assets, the above-median indicator, and the composite z-score – each of which displays the same E1-positive, E3-negative pattern.

Table 21: Size Effect Across Alternative Measures

	E1	E2	E3
Panel A: Univariate			
log(Market cap)	0.0054***	-0.0012	-0.0059**
	(0.0018)	(0.0018)	(0.0026)
N	272	273	269
R^2	0.038	0.002	0.024

Panel B: Alternative measures (E1 vs. E3 sign flip)

	E1 β	E1 p	E3 β	E3 p	Sign Flip
log(Market cap)	+0.00540	(0.003)	−0.00585	(0.024)	Yes
log(Assets)	+0.00309	(0.074)	−0.00542	(0.036)	Yes
Above median	+0.02492	(0.000)	−0.01751	(0.074)	Yes
Top decile					
z-size (reversed)	−0.00980	(0.018)	+0.01523	(0.013)	Yes

Notes: Panel A: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around the event date, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Panel B: Univariate regressions per measure. z_size sign reversed (higher = smaller). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.5 Continuous Size Measure

Table 22 re-estimates the horse race specification with the continuous $\ln(\text{MktCap})_i$ measure in place of the binary above-median indicator. The size coefficient enters positively throughout, though with wider confidence intervals than the binary measure: 0.0032 in the joint specification, where it is marginally significant, and 0.0035 in the governance-augmented specification, where it reaches significance at the 5% level. Those wider standard errors trace to the concavity of the size–CAR relationship documented in Figure 3, where the gradient is steep below the median and flat above it, which favours a threshold specification over a log-linear one. All remaining coefficients are virtually unchanged between the continuous and binary specifications.

Table 22: Horse Race with Continuous Size Measure (E1)

	(1)	(2)	(3)	(4)	(5)
log(Market cap)	0.0054***		0.0032*	0.0036	0.0035**
	(0.0018)		(0.0017)	(0.0023)	(0.0017)
Friction count		-0.0034***	-0.0032***	-0.0032**	-0.0031**
		(0.0013)	(0.0012)	(0.0015)	(0.0013)
Watchlist			0.0505	0.0560	0.0498
			(0.0327)	(0.0390)	(0.0330)
Illiquidity (z-Amihud)			-0.0017	-0.0014	-0.0010
			(0.0095)	(0.0107)	(0.0095)
Catalist			-0.0114	-0.0018	-0.0074
			(0.0124)	(0.0179)	(0.0127)
Independence share					-0.0184
					(0.0249)
No CEO duality					0.0091
					(0.0101)
Sector FE	No	No	No	Yes	No
N	272	274	271	243	260
R^2	0.038	0.049	0.087	0.195	0.083

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1 (February 21, 2025), estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Same specification as Table 18 but using continuous $\ln(\text{MktCap})_i$ instead of the binary above-median indicator. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.6 Generated-Regressor Bootstrap

Because the dependent variable is itself an estimate, the second-stage standard errors may understate true uncertainty,[115] and we address that concern with a two-stage bootstrap of $B = 2{,}000$ replications over a pre-E1 estimation window of $(-250, -30)$. The first stage applies a moving-block bootstrap (block length 5 trading days) to each firm's joint factor-and-return time series, re-estimating Maynes–Rumsey factor loadings and recomputing CARs at the original event window. The second stage applies a pairs bootstrap over firms for the E1 horse race and a cluster-pairs bootstrap, with clusters defined by stock code, for the stacked E1–E3 specification, so that the bootstrap distribution absorbs both the first-stage estimation noise and the cross-sectional residual variation.

Table 23 sets the bootstrap standard errors and 95% percentile intervals against the conventional

[115] Pagan (n 94)

HC1 (E1 horse race) and CR1 (stacked reversal) baselines. The two headline coefficients remain statistically significant under bootstrap inference: the E1 above-median size effect ($p = 0.007$, SE wider by 27%) and the E1-to-E3 reversal γ_2 ($p = 0.001$, SE wider by 17%) each remain significant at the 1% level. The friction coefficient, by contrast, attenuates to marginal significance ($p = 0.054$), which exposes first-stage noise that HC1 inference does not register. The FF3 watch-list coefficient is not statistically significant under bootstrap inference ($p = 0.30$), an outcome consistent with the five-firm treatment cell and the SMB-absorption issue already discussed in the factor-robustness appendix.

Table 23: Generated-Regressor Bootstrap ($B = 2{,}000$)

	Coef.	Conv. SE	Conv. p	Boot SE	Boot 95% CI	Boot p
AboveMedian$_i$ at E1 (Table 10)	0.0227***	(0.0062)	0.000	0.0079***	[+0.0087, +0.0404]	0.007
FricCount$_i$ at E1 (Table 10)	−0.0031**	(0.0012)	0.011	0.0016*	[−0.0065, −0.0002]	0.054
Watchlist$_i$ at E1 (Table 10)	0.0524*	(0.0316)	0.097	0.0543	[−0.0536, +0.1644]	0.304
AboveMedian$_i$ $\times \mathbb{1}_{e=\mathrm{E3}}$ reversal (Table 8)	−0.0424***	(0.0116)	0.000	0.0136***	[−0.0643, −0.0108]	0.001

Notes: Two-stage bootstrap with $B = 2{,}000$ replications. First stage: moving-block bootstrap (block length 5 trading days) of each firm's joint factor-and-return time series within the pre-E1 estimation window $(-250, -30)$; Maynes–Rumsey factor loadings re-estimated and CARs recomputed at the original $(-1, +3)$ event window. Second stage: pairs bootstrap over firms (E1 horse race) or cluster-pairs bootstrap on stock code (stacked E1–E3 specification). "Conv. SE" is HC1 (E1 horse race) or CR1 clustered on stock code (stacked reversal). "Conv. p" is the two-sided asymptotic p-value implied by the conventional standard error; stars on the coefficient column follow this p-value. "Boot 95% CI" is the 2.5%–97.5% percentile interval of the bootstrap distribution. "Boot p" is the two-sided percentile p-value against H_0: coefficient = 0; stars on the bootstrap SE column follow this p-value. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.7 Catalist Board-Tier Effect

The Catalist board-tier coefficient reported in preliminary analyses, significant at the 1% level (−3.0 percentage points), turns out to be an artefact of sample composition. The initial sample included non-equity stock codes (warrants, structured products, daily leverage certificates), all classified as Mainboard, whose positive mean CARs at E1 inflated the Mainboard reference group and exaggerated the apparent Catalist underperformance. Table 24 reports the equity-only results. Under CAPM the bare Catalist coefficient is −2.82 percentage points, significant at the 1% level (Panel A), and the FF3 model, which adds the SMB factor, attenuates it to −2.44 percentage points, significant at the 5% level; adding the above-median size indicator carries it further, to −1.23 percentage points and insignificance. Since Catalist firms are on average smaller than Mainboard firms, the board-tier variable mainly proxies for size.

Table 24: Catalist Board-Tier Effect

Panel A: Catalist coefficient by model and controls (E1)

	FF3		Market Model	
	Bare	+Size	Bare	+Size
Catalist	-0.0244**	-0.0123	-0.0282***	-0.0137
	(0.0121)	(0.0132)	(0.0106)	(0.0117)
Above median		0.0214***		0.0257***
		(0.0071)		(0.0068)
N	274	274	274	274
R^2	0.025	0.057	0.037	0.087

Panel B: Board means by event (FF3 CAR$(-1, +3)$ %)

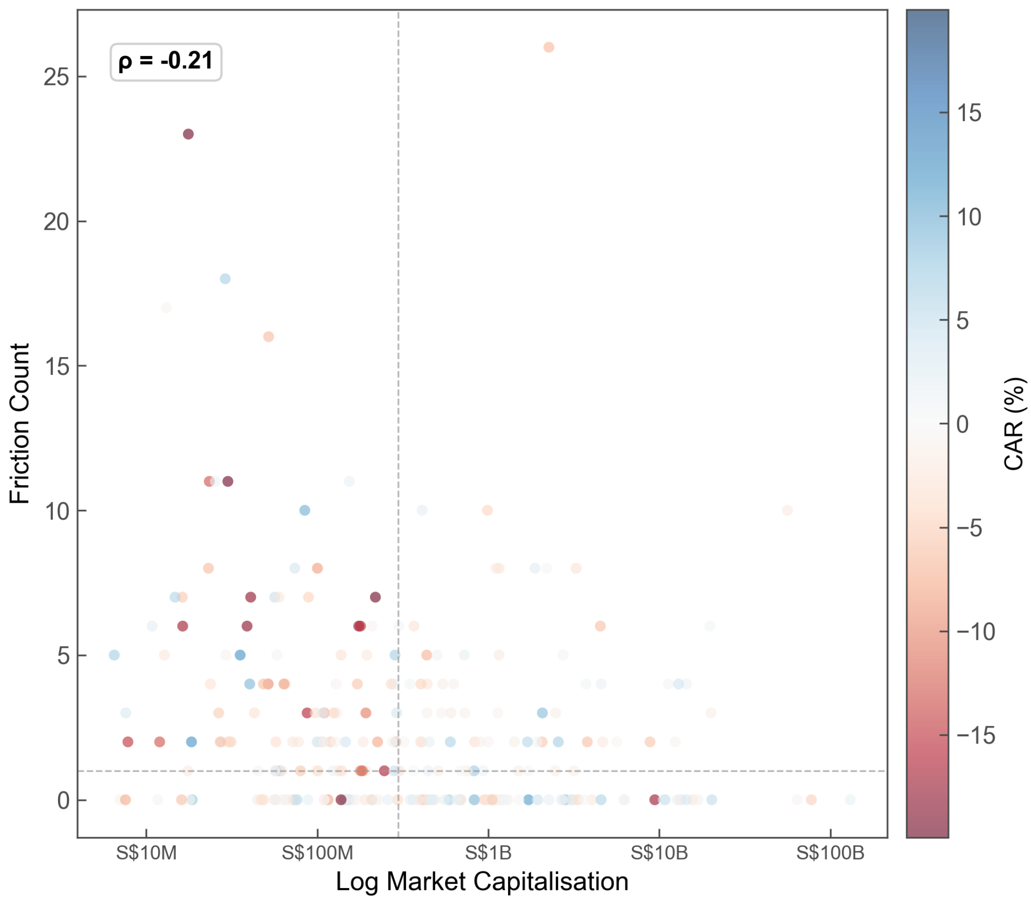
	E1	E2	E3
Mainboard	-0.95	-0.23	+2.65
	(0.33)	(0.28)	(0.44)
Catalist	-3.39	+0.55	+5.61
	(1.18)	(1.30)	(1.66)
Diff (Cat−MB)	-2.44*	+0.77	+2.96*
	(-2.00)	(0.58)	(1.72)

Notes: Panel A: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1. FF3: Fama–French three-factor model (Asia Pacific ex Japan). Market Model: CAPM with SGX equal-weighted index. The FF3 size factor (SMB) absorbs most of the board effect. Panel B: Mean CARs (%) by board tier (equity-only sample). Conventional standard errors of the mean $(s/\sqrt{n})$ are reported in parentheses for board means; significance is assessed by the Student t-test. Welch t-statistic in parentheses for the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.8 Size–Friction Correlation

Figure 7 plots log market capitalisation against the regulatory friction count. The moderate negative correlation ($\rho = -0.21$) indicates that larger firms tend to accumulate fewer friction events, which is the overlap between the capital and information channels that one would expect. That correlation is nonetheless weak enough to leave the two variables distinct for the joint estimation in the horse-race regressions of Table 18.

Figure 7: Scatter plot of log market capitalisation against regulatory friction count. The negative correlation (Pearson $\rho = -0.21$) confirms that size and friction are empirically related but not collinear.

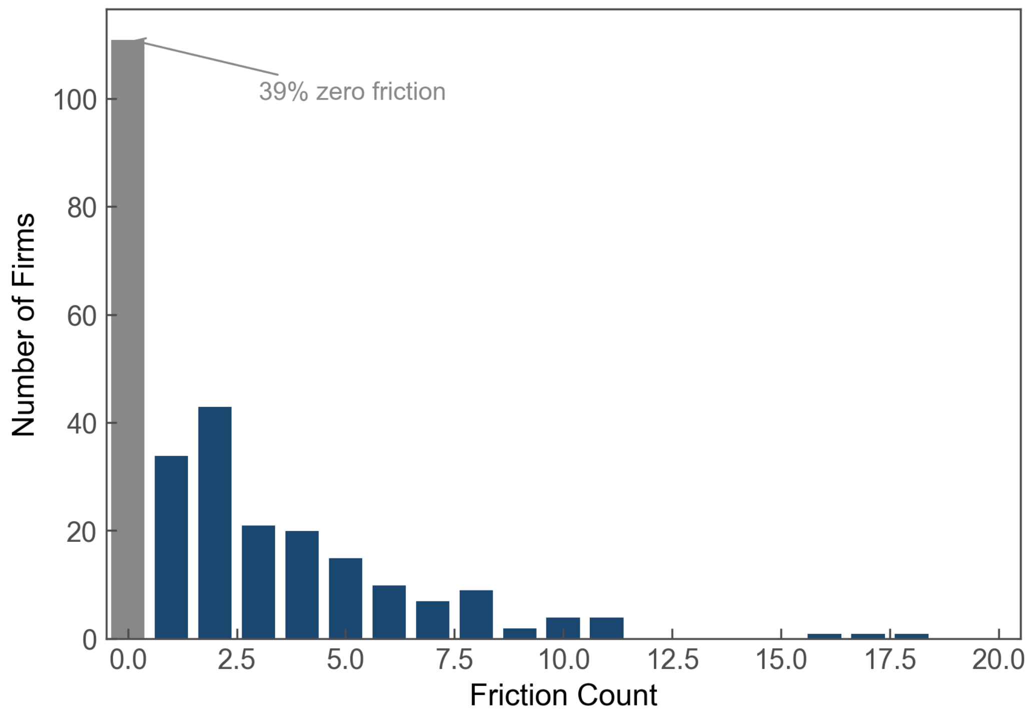


Note: Each point represents a firm at E1. Colour indicates FF3 CAR($-1, +3$), winsorised at the 1st and 99th percentiles. Dashed lines mark sample medians. ρ denotes the Pearson correlation between log market capitalisation and total friction count ($\rho = -0.21$), confirming that the two channels are not collinear.

A.3 Supplementary Figures

A.3.1 Friction Distribution

Figure 8: Distribution of regulatory friction counts across the SGX equity panel. The majority of firms have zero or one friction event; the right tail is driven by firms with repeated queries, halts, and suspensions.



Note: Distribution of total friction event count across the 335 equity-only sample firms. Friction events include SGX queries, trading halts, and suspensions over August 2022 to August 2025.

Figure 8 displays the heavily right-skewed friction distribution that underlies the estimates in Table 10, which confirms that the linear specification is driven primarily by the contrast between zero-friction and moderate-friction firms.

A.4 Additional Event and Cross-Sectional Results

A.4.1 E0: Review Group Formation

The Review Group formation announcement on August 2, 2024 (E0) produced a strongly negative mean CAR of −2.15%, significant at the 1% level, with only 30.8% of firms posting positive returns, and the cross-sectional regressions yield significant size and friction coefficients in line with the patterns documented for E1. The event window, however, coincides with the late-July to early-August 2024 JPY carry trade unwind, which generated large negative returns across Asian equity markets. Augmenting the market model with a JPY factor (FF4-JPY) makes the REIT effect and several other cross-sectional patterns disappear, which indicates that the E0 results are confounded by the carry trade unwinding. We therefore treat E0 as uninformative for hypothesis testing and rely on E1 as the primary announcement event.

A.4.2 E2 and E3 Detailed Results

The E2 consultation event (May 15, 2025) produced a null mean CAR of −0.11% ($p = 0.721$), and no cross-sectional variable reaches significance at conventional levels. This null is consistent with the consultation having been fully anticipated once E1 announced that detailed proposals would follow "by mid-2025."

E3 (July 21, 2025) produced a uniformly positive mean CAR of 3.12%, significant at the 1% level, with 70.0% of firms posting positive returns. Unlike E1, the continuous size measure reverses sign here, to a value significant at the 5% level (−0.006), while friction count is economically zero ($p > 0.55$). The capital deployment raised abnormal returns across the equity panel, though smaller firms benefited disproportionately. The reversal of the size gradient at E3, set against its positive sign at E1, supplies the central evidence for the intermediation gap hypothesis: once mandate terms specifying small- and mid-cap allocation accompanied the capital deployment, the market unwound the size-based differentiation it had priced in at the announcement.

A.4.3 E5: Final Report

The final report event (E5, November 19, 2025) produced a positive mean CAR of 0.62%, significant at the 1% level, with 60.5% of firms posting positive returns. In the cross-section the above-median size indicator is positive and significant at the 1% level (0.015), and illiquidity carries a positive sign significant at the 5% level (0.007), the opposite direction from E1. Friction count and watch-list status are both insignificant.

A.4.4 Governance Variables

Table 25 reports univariate governance regressions across E1, E2, E3, and E4. Board independence share, non-busy board membership, and turnover rate are insignificant at all four events, and CEO-chair separation is insignificant at E1, E2, and E4, reaching marginal significance only at E3 (−0.022). Fresh independence – whether the board appointed at least one new independent director in the preceding fiscal year – is positive and significant at the 1% level at E3 (0.092) and at the 5% level at E4 (0.019), which suggests that firms with recently refreshed boards benefited disproportionately from the capital deployment and the deregulation implementation.

Table 25: Governance Variables Across Events

	E1	E2	E3	E4
Independence share	0.0038	-0.0240	-0.0113	-0.0131
	(0.0260)	(0.0243)	(0.0375)	(0.0374)
No CEO duality	0.0108	0.0087	-0.0222*	0.0015
	(0.0104)	(0.0092)	(0.0126)	(0.0096)
Non-busy board	-0.0264	-0.0016	-0.0469	-0.0194
	(0.0255)	(0.0252)	(0.0409)	(0.0307)
Fresh independence	-0.0131	0.0019	0.0919***	0.0192**
	(0.0101)	(0.0125)	(0.0113)	(0.0081)
Turnover rate	-0.0046	-0.0020	-0.0019	-0.0002
	(0.0029)	(0.0026)	(0.0053)	(0.0030)

Notes: Univariate OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Governance variables from director data. E4 governance merged from E1 cross-section. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4.5 Broker/Institutional Ownership

Singapore's nominee and custody structure holds substantial shareholdings through intermediary accounts, which complicates any reading of the register. We decompose the top-20 shareholder register into four categories: insider (directors and substantial shareholders), nominee (bank nominee accounts), broker (retail securities brokers), and institutional (fund managers and sovereign wealth funds). Although nominee accounts hold shares on behalf of any investor type, broker accounts predominantly serve retail investors who maintain securities accounts with their brokerage. We therefore use *Broker %* – the share of top-20 holdings registered through retail securities brokers – as a proxy for retail-dominated ownership and, inversely, for the absence of the institutional infrastructure that facilitates price discovery around information events.

Table 26 estimates stepwise OLS regressions of E1 abnormal returns on broker ownership. Univariately, a one percentage point increase in broker ownership is associated with a -0.0007 lower CAR, significant at the 1% level (column 1). Adding size controls attenuates the coefficient to -0.0006, significant at the 5% level (column 2), and adding friction controls in turn yields -0.0005, again significant at the 5% level (column 3). Even in the most saturated specification, which adds financial controls, illiquidity, and Catalist (column 5), the coefficient holds at -0.0005 and marginal significance, which points to a capital channel operating independently of firm size and market frictions.

Table 27 sharpens this pattern through broker-ownership quartile sorts. At E1, mean CARs decline monotonically from -0.36% in Q1 (lowest broker ownership) to -2.37% in Q4 (highest), so the $Q4 - Q1$ spread reaches -2.00 pp and is marginally significant. At E3, the ranking inverts. All four quartiles earn positive mean CARs, and the $Q4 - Q1$ spread flips sign to 1.70 pp, though it is now statistically indistinguishable from zero. This E1-to-E3 reversal in the broker-ownership gradient tracks the E1-to-E3 size-gradient reversal documented in the main results, and it is consistent with retail-dominated firms being discounted on the announcement of deregulatory measures and then re-rated once the capital deployment terms were disclosed.

Table 28 asks whether post-reform ownership changes line up with the policy's intended direction. Had the programme channelled capital toward small and mid-cap firms, nominee holdings – the primary custodial vehicle for institutional investment – should have risen disproportionately for smaller firms. The observed pattern runs the other way. Controlling for FY2024 nominee levels, larger firms exhibit higher FY2025 nominee ownership (coefficient on log(MktCap) = 1.25, significant at the 5% level; column 1), and Catalist-listed firms show a 3.5 percentage point decline in broker holdings relative to Mainboard peers, also significant at the 5% level (column 2). These ownership patterns align with the announcement-return evidence that the capital channel favoured larger, more liquid firms, although they do not by themselves isolate the causal effect of the EQDP on ownership structure.

The broker ownership subsample covers 205 of 274 E1 firms (75%); the excluded firms are marginally smaller ($p = 0.082$) and overwhelmingly Catalist-listed, a composition difference significant at the 1% level (97% vs 81%). CARs and friction counts, by contrast, do not differ significantly across the two subsamples. The ownership rebalancing sample is narrower still, covering the 172 firms with top-20 shareholder data in both FY2024 and FY2025. The results that follow should be read with these coverage limits in mind.

Table 26: Broker Ownership and E1 Abnormal Returns

	(1)	(2)	(3)	(4)	(5)
Panel A: Broker ownership and E1 abnormal returns					
Broker %	-0.0007^{***}	-0.0006^{**}	-0.0005^{**}	-0.0006^{**}	-0.0005^{*}
	(0.0002)	(0.0003)	(0.0002)	(0.0002)	(0.0003)
Log(Mktcap)		0.0036	0.0023	0.0027	0.0011
		(0.0023)	(0.0022)	(0.0022)	(0.0026)
Friction count			-0.0033^{*}	-0.0036^{**}	-0.0037^{**}
			(0.0018)	(0.0017)	(0.0018)
Watchlist				0.0699^{*}	0.0628
				(0.0385)	(0.0419)
Illiquidity					0.0027
					(0.0095)
Catalist					−0.0057
					(0.0139)
Financial controls	No	No	No	No	Yes
N	205	204	204	204	192
R^2	0.046	0.059	0.094	0.115	0.147

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is FF3 CAR(−1, +3), estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Broker % is the share of top-20 holdings registered through retail securities brokers. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 27: Broker Ownership Quartile Sorts

	E1: February 2025		E3: July 2025	
	Mean	Median	Mean	Median
Q1 (lowest)	-0.36	-0.12	$+1.97^{**}$	$+0.76$
	(0.66)		(0.93)	
Q2	-0.45	-0.54	$+3.16^{***}$	$+1.91$
	(0.57)		(0.97)	
Q3	-1.75^{*}	-1.52	$+4.98^{***}$	$+3.24$
	(0.97)		(1.35)	
Q4 (highest)	-2.37^{***}	-1.94	$+3.68^{***}$	$+2.85$
	(0.86)		(1.36)	
$Q4 - Q1$	-2.00^{*}		$+1.70$	
	(-1.85)		(1.03)	
N	205		201	

Notes: Mean and median CARs (%) by broker-ownership quartile for E1 (21 February 2025) and E3 (21 July 2025). Conventional standard errors of the mean ($s/\sqrt{n}$) are reported in parentheses; significance is assessed by the Student t-test. Stars on mean CARs denote the significance of a one-sample t-test against zero. The $Q4 - Q1$ row reports the difference in means; Welch t-statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 28: Post-E3 Ownership Rebalancing

	(1)	(2)	(3)
Nominee % (FY2024)	0.8369***		
	(0.0537)		
Log(Mktcap)	1.2515**	−0.2459	
	(0.5484)	(0.2659)	
Catalist	0.2180	−3.5228**	
	(1.6345)	(1.5889)	
Medium			−1.0171
			(1.6969)
Large			−2.8088
			(2.0851)
N	172	172	172
R^2	0.815	0.025	0.015
Large-cap Δ Top-1 %	−2.92** [$t = -2.22$, $p = 0.031$]		

Notes: OLS with HC1 robust standard errors in parentheses. Dependent variables are FY2025 nominee ownership (column 1), the FY2024-to-FY2025 change in broker ownership (column 2), and the FY2024-to-FY2025 change in top-1 shareholder ownership (column 3), estimated on the subsample of firms with top-20 shareholder data in both fiscal years. Nominee % (FY2024) is the lagged dependent variable. Medium and Large are size-tercile indicators (omitted category: Small). The final row reports the mean change in top-1 shareholder ownership for large-cap firms. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4.6 Realised Dollar Volume Across Reform Sub-Periods

The paper's central identification is cross-sectional within announcement windows, which makes it an expectational claim about how the market repriced firms at each reform event. This appendix supplements that identification with a descriptive analysis of realised trading volume across the reform timeline, decomposing the post-E1 period into four sub-periods and asking whether the relative size gradient in realised dollar volume tracks the announcement-window CAR pattern. The exercise is descriptive for two reasons: each post-E1 sub-period spans weeks to months, over which many shocks unrelated to the reform occur, and raw dollar volume cannot be netted against a factor-model benchmark as it is in the estimation of CARs. The single-date cross-sectional design that identifies the main results therefore cannot be reproduced over these extended sub-periods.

The sample is a balanced panel of 260 SGX-listed equity issuers (128 above-median and 131 below-median at the E1 cross-section), each contributing at least three trading days in every sub-period. The reference baseline is Pre-E1 (January 2, 2024 to February 14, 2025), which matches the estimation window used in the paper's CAR specifications. Four post-E1 sub-periods are compared against this baseline:

1. E1→E3: February 21 to July 20, 2025 (post-reform announcement, pre-mandate disclosure).
2. E3→E4: July 21 to October 28, 2025 (EQDP active deployment, pre-implementation of E4 reforms).
3. E4→E5: October 29 to November 18, 2025 (post-watch-list abolition, anticipating E5 structural reforms).
4. Post-E5: November 19 to December 31, 2025 (post-implementation of E5).

Splitting E3→E4 from E4→E5 separates EQDP active deployment from the deregulation and structural-reform packages introduced at E4 and E5. From E3 (July 21) the named managers are ostensibly engaged in active deployment, whereas the E4→E5 window absorbs both the SGX RegCo deregulation implemented at E4 and any market anticipation of the E5 structural package, both of which Section 3 details. The E3→E4 sub-period is therefore the one window that isolates intermediated deployment from subsequent reform.

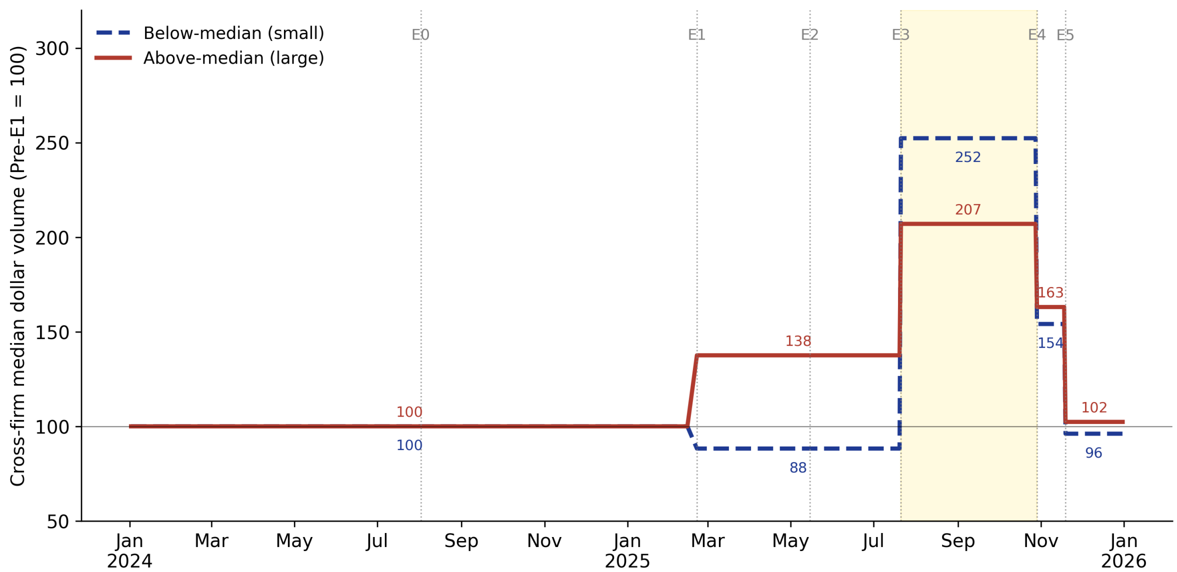
Table 29 reports cross-firm median dollar volume per firm-day for each group and sub-period, together with percent changes from Pre-E1, and Figure 9 indexes each series to its Pre-E1 baseline of 100 and plots the period-level statistic as a step function. Through E1→E3 the below-median series sits below baseline while the above-median series rises. Both groups then jump sharply at E3→E4, where the below-median series reaches an index of 252 against 207 for the above-median group; the gains attenuate through E4→E5 and Post-E5, and both groups return to within a few percent of baseline by the end of December 2025.

Table 29: Cross-Firm Median Dollar Volume per Firm-Day, by Sub-Period

Period	Below-median (small)		Above-median (large)	
	Median $/firm-day	Δ% vs Pre-E1	Median $/firm-day	Δ% vs Pre-E1
Pre-E1 (baseline)	$40,459	—	$1,027,699	—
E1→E3	$35,708	−11.7%	$1,413,671	+37.6%
E3→E4	$102,086	+152.3%	$2,127,791	+107.0%
E4→E5	$62,353	+54.1%	$1,675,995	+63.1%
Post-E5	$38,885	−3.9%	$1,051,942	+2.4%

Notes: Balanced panel of 260 SGX-listed equity issuers (128 above-median, 131 below-median at the E1 cross-section), each with at least three trading days in every sub-period. Each cell reports the cross-firm median of within-firm mean dollar volume per trading day in the indicated sub-period, with percent change from the Pre-E1 baseline. Pre-E1 spans January 2, 2024 to February 14, 2025 (the CAR estimation window).

Figure 9: Rolling Median Dollar Volume by Group



Notes: Cross-firm median of within-firm period mean dollar volume by group, indexed to Pre-E1 = 100. Each step corresponds to the period-level statistic reported in Table 29. Dotted vertical lines mark the reform events. The shaded region demarcates the E3→E4 sub-period, the only window in the timeline that isolates EQDP active deployment from subsequent reform implementation. Sample: 260 SGX-listed equity issuers.

The E3→E4 sub-period is the only post-E1 window in which below-median firms grew faster than above-median firms in proportional terms; in E1→E3, E4→E5, and Post-E5, the proportional ranking reverses, and above-median firms grow faster.[116] This pattern is consistent with the EQDP active-deployment phase generating proportional trading-volume growth that favours small-cap firms.

The size-gradient widening evident at E4→E5 and Post-E5 surfaces only once the subsequent deregulation and structural-reform package enters the window. Three components of that package carry a mechanical large-cap incidence: the SGX-NASDAQ dual-listing bridge, by easing a primary listing on both boards, carries fixed costs that favour larger issuers; the board-lot reduction above S$10 lowers minimum trade size only for higher-priced stocks, which cluster in larger capitalisations; and DMM rebates raise the expected per-quote revenue of market making in the firms where it is already profitable, predominantly mid-caps. Each of these channels operates outside the paper's capital and information channels.

The descriptive evidence supports two refinements to the headline narrative. First, the mandate-specificity reversal at E3 corresponds to a cross-sectional repricing of expectations as well as a real volume response across the size distribution during the deployment phase. Second, the post-E5 widening of the size gap in realised trading is best attributed to the layered structural

[116] A within-firm panel regression of daily log(1 + shares traded) on AboveMedian × sub-period interactions, with firm and date fixed effects and firm-clustered standard errors, yields coefficients of +0.25 ($p = 0.013$) at E1→E3, +0.13 ($p = 0.45$) at E3→E4, +0.56 ($p = 0.026$) at E4→E5, and +1.18 ($p < 0.001$) Post-E5; a Wald test rejects equality of the E3→E4 and E4→E5 coefficients ($\Delta\hat{\beta} = -0.42$, $p = 0.016$). A continuous-size specification replacing the binary indicator with log market capitalisation yields qualitatively identical results.

reforms described above, which intervene after deployment, rather than to any failure of the capital channel.

Several caveats qualify this descriptive evidence. The E3→E4 result isolates the EQDP active-deployment window from subsequent reform implementation, yet coincident factors – sector cycles, generalised post-mandate sentiment, and unobserved investor positioning – cannot be ruled out. The two later windows are also thin: E4→E5 spans only about 15 trading days and Post-E5 about 30, the latter overlapping SGX year-end seasonality, so the proportional rankings there rest on few observations and the persistence of the size-gap widening cannot be assessed within this sample. A final limitation is more basic, since the analysis cannot resolve whether buy-and-hold deployment by EQDP managers materially limited the volume footprint of intermediated capital: the available institutional-ownership disclosures reflect fiscal-year-ends that predate E3.